

Managerial Decision Making
Business 38802 (AXP 25)
Autumn 2025
George Wu

Class 5

“Nudges”/Improving Decision
Processes

Course Outline

Monday, November 3	Introduction to Decision Making
Tuesday, November 4	Judgment under Uncertainty I (Heuristics and biases)
Wednesday, November 5*	Judgment under Uncertainty II (The difficulty of learning)
Friday, November 7	Risky Decision Making
Saturday, November 8	Nudges/Improving Decision Processes
January 5	Final (More next)

Last Debrief Note

- Optional!

- ❑ Best 4 of 5
- ❑ Due: Tuesday, November 11, 5:00 PM

Gawande again...translated



“The core predicament of business – the thing that makes being a decision maker so wrenching, being an executive so difficult, and being a part of a society that suffers the consequences of poor decision making – is uncertainty ... Business’ ground state is uncertainty. And wisdom – for managers of all ranks – is defined by how one copes with it.”

- Atul Gawande (2002), *Complications* (“The case of the red leg”)

What did we learn yesterday?

- Managing uncertainty involves *appropriate* risk attitudes (more today)
- Regret
 - Class 1: Asymmetries in evaluation by your boss + outcome bias (hindsight) lead you to be risk-averse
 - Regret means that you are also your own evaluator
- Prospect Theory
 - People are loss averse (**2** is the magic number!)
 - Explains **lots** of stuff, including why change is hard
 - Probabilities are distorted
 - Risk aversion for most gains/risk seeking for most losses
 - Reverse for small probabilities (lottery tickets/insurance)

What did we learn yesterday?

- Mental accounting

- Machinery of prospect theory (e.g., segregating gains)
 - Early gains (0 to 1,000) are more attractive than later gains (75,000 to 76,000) (bonus > increase in salary)
 - Reducing a loss is more attractive than creating a gain (paying off loan > bonus)
 - Reducing a gain is more attractive than creating a loss (vacation is more palatable if you have a “mental” pot of money)
- Sunk Costs

- Policy-based decision making

- A decision approached one-at-a-time leads people to be risk-averse

Today's Objectives

- Goals: Nudges/Improving Decision Processes
 - Understanding the basics of “choice architecture”
 - Thinking smarter about decisions (the “art and science” of decision analysis)
 - Challenges to improving organizational decision making
 - psychological and organizational issues
- Outline
 - Discussion of Express Scripts
 - Lecture on choice architecture
 - An analytical and psychological framework for good decision making
 - Data: Ratings and best/worst practices
 - How you and your organization can make better decisions
- Note: Lots of hidden slides (included in Canvas)

Decision Making @ Google

“The most important thing to do is to have quick decisions. And you’ll make some mistakes, but you need decision-making...I cannot tell you how many people have told me that at Google decisions are made today quickly in almost every case, even at our current scale... Most large corporations have too many lawyers, too many decision-makers, unclear owners, and things congeal, they occur very slowly...Even if it’s the wrong decision, a quick decision is better than almost every case.”

Eric Schmidt



Decision Making @ Chewy

“The first principle of developing good judgment is to gain varied experiences and to live in a feedback loop of continuous evaluation and improvement. It is inevitable that you will find yourself in situations where you do not have perfect information. This is where risk-taking comes into play.

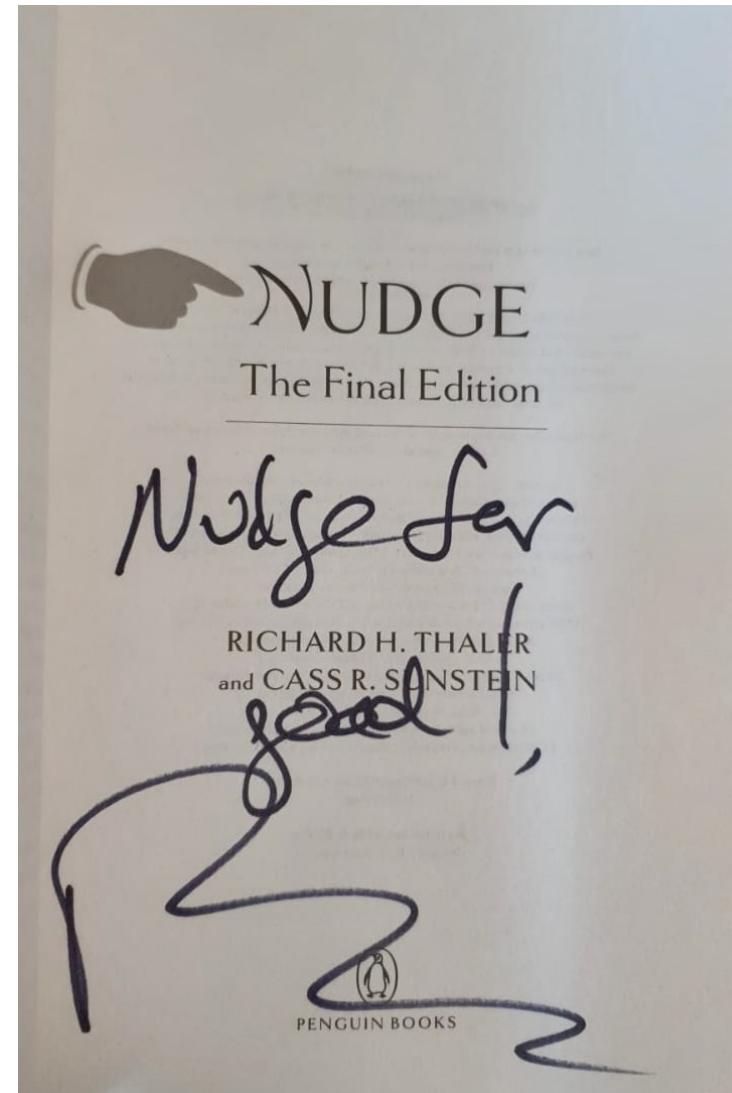
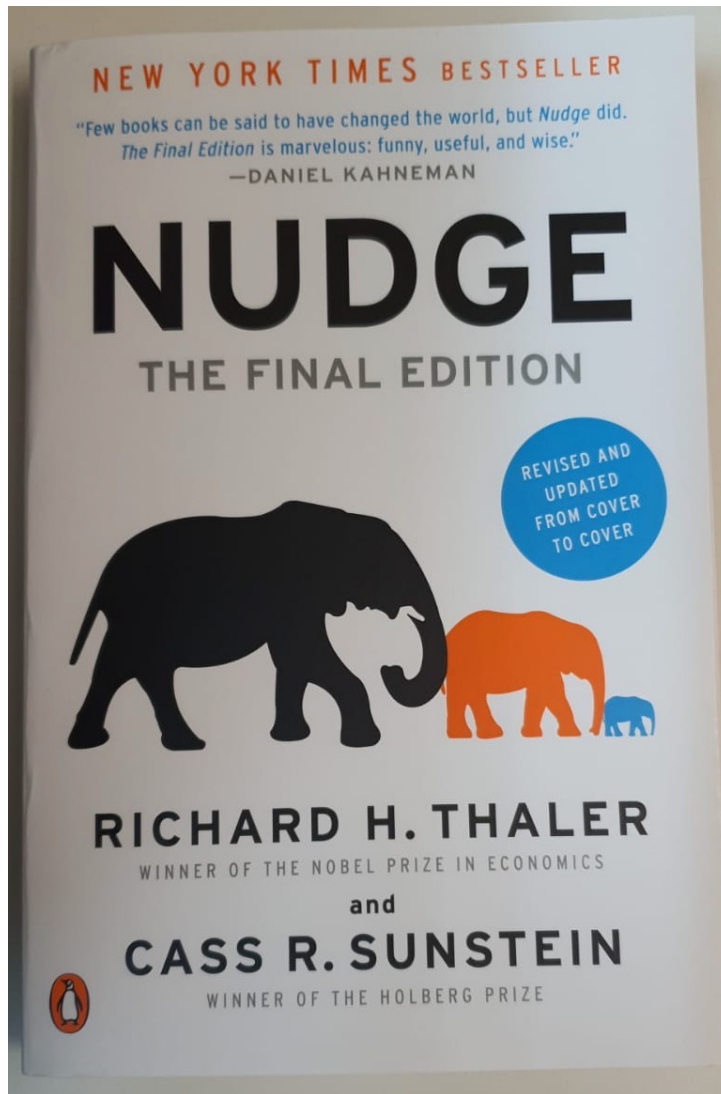
...

I knew I didn't have the full picture, but I was comfortable with the unknown, with accepting that Chewy might fail, and I might fail.”

Sumit Singh '14



Back to this!

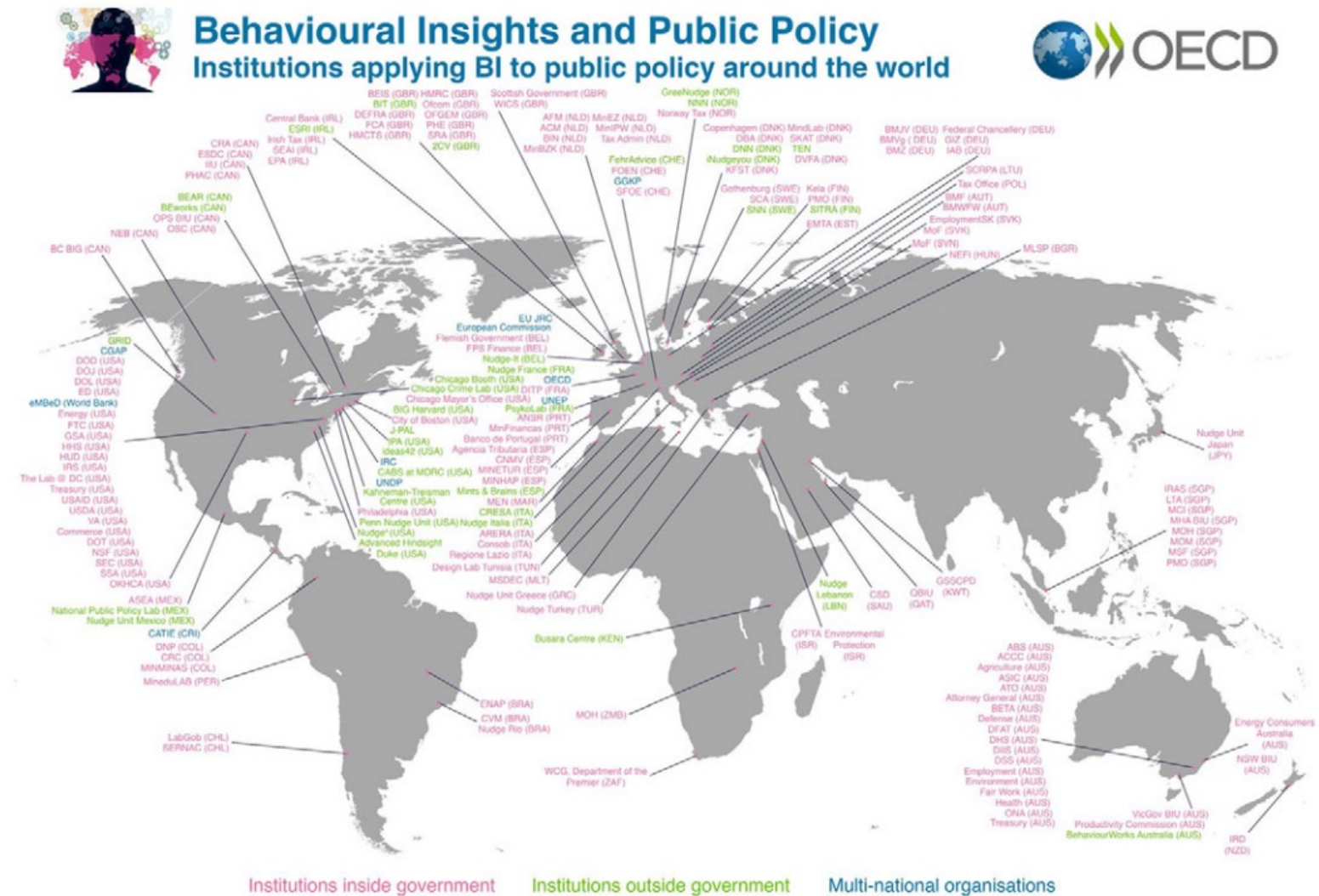


Thaler Nobel Prize

Richard Thaler's contributions have built a bridge between the economic and psychological analyses of individual decision-making. His empirical findings and theoretical insights have been instrumental in creating the new and rapidly expanding field of behavioural economics, which has had a profound impact on many areas of economic research and policy.

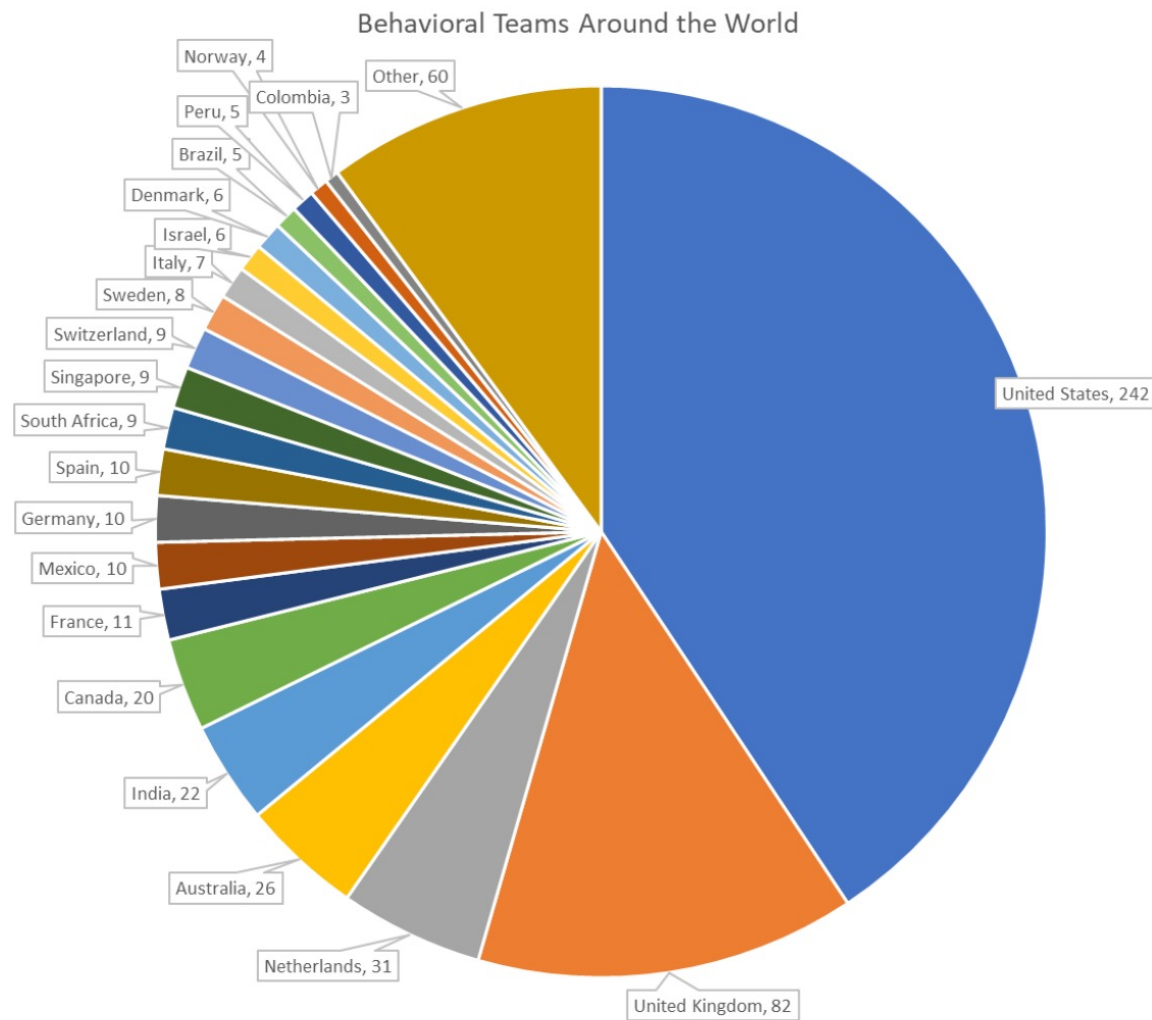


Nudge Units



Source: OECD Research (2018)

Nudge Units

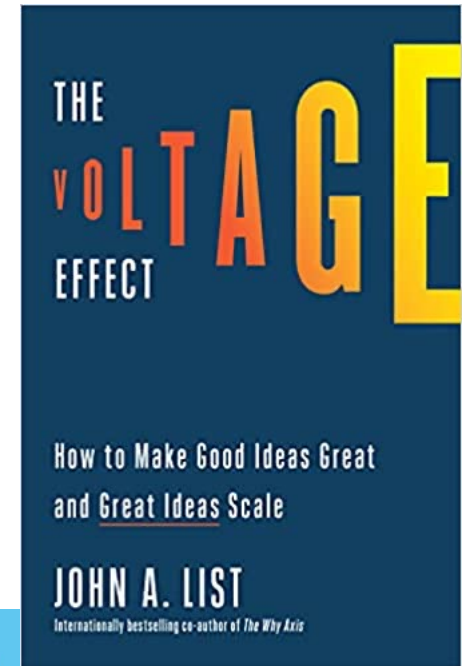
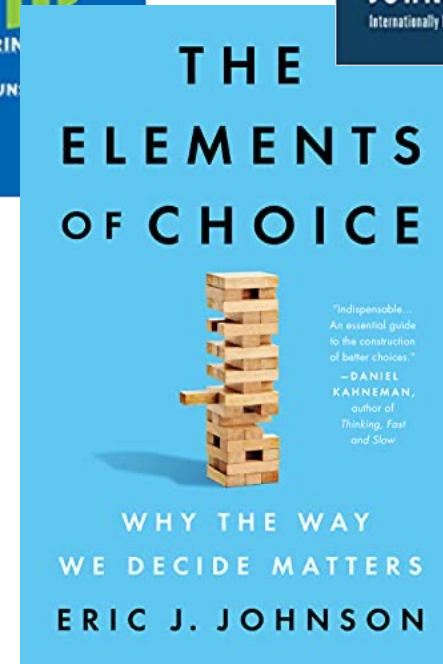
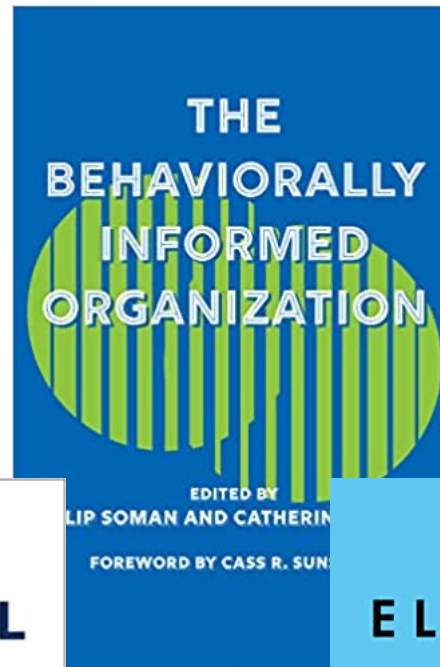
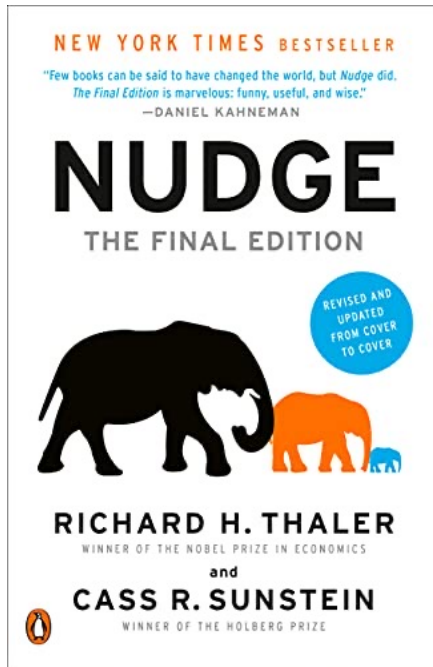


Source:
Wendel
(2020)

“Behavioral Economics” Recipe Book

- Use principles of behavioral science (this class + Managerial Psychology)
- Design choice architecture
- Experiment!
- Iterate!

Lots of Books Out There



Components of Nudges

- What is choice architecture?
- Ethos of experimentation
- Understanding barriers to good choices

Choice Architecture

A *choice architect* is anyone who designs the environment in which people make choices

- ❑ Restaurant Menus
- ❑ Store layouts
- ❑ Curriculum
- ❑ Websites
- ❑ Apps
- ❑ HR benefits booklets (ugh!)
- ❑ Informational environments



Identify a decision in your firm in which some aspect of choice architecture has a dramatic effect on decisions. Do you like the outcome? If not, what should you do?

Example: Corporate Cafeteria

■ Decisions

- ❑ Menu
 - Taste, health, variety
- ❑ Prices of menu items
- ❑ Location of items
- ❑ Information (calories, allergies...)

■ Objectives

- ❑ Maximize profit
- ❑ Get employees to eat healthy
- ❑ Minimize/maximize time in cafeteria
- ❑ Maximize productivity
- ❑ Sell your uncle's energy drink

■ What should you do?

- ❑ Where do you put the salad bar?



Experimentation

In the real world, we observe:

- Usual state of affairs: A decision outcome, along with tons of variables (decisions we made + other factors)
- Our Goal: Determine whether a decision or a factor caused a change in an outcome
 - ▣ Example: Toro

Main point and a huge challenge: How do we know that sales growth was due to our promotion and wasn't due to something else?

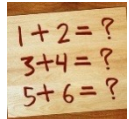


Identify a setting in which your organization could experiment. What are the opportunities? Challenges?

Experiments 101

An experiment consists of:

- A set of (theory-driven) treatments or conditions
- *Random* assignment of people to treatment (best if people do not know they are in an experiment!)
- Measurement of relevant outcomes
- Statistical procedure to determine if result is “significant” (require a sufficiently large sample)



How would you conduct an experimental test of S'No Risk?

Why don't we make good choices?

- Everything that we talked about in this class!
 - ❑ Judgment biases (e.g., availability and anchoring)
 - ❑ Framing
 - ❑ Analysis Paralysis
- Much of what was discussed in Managerial Psychology!
- Lots of other stuff
 - ❑ Lack of knowledge
 - ❑ People are myopic
 - ❑ People are busy
 - ❑ People are conflicted
 - ❑ People are overwhelmed

The U.K. Behavioral Insights Team

Also known as “The Nudge Unit”



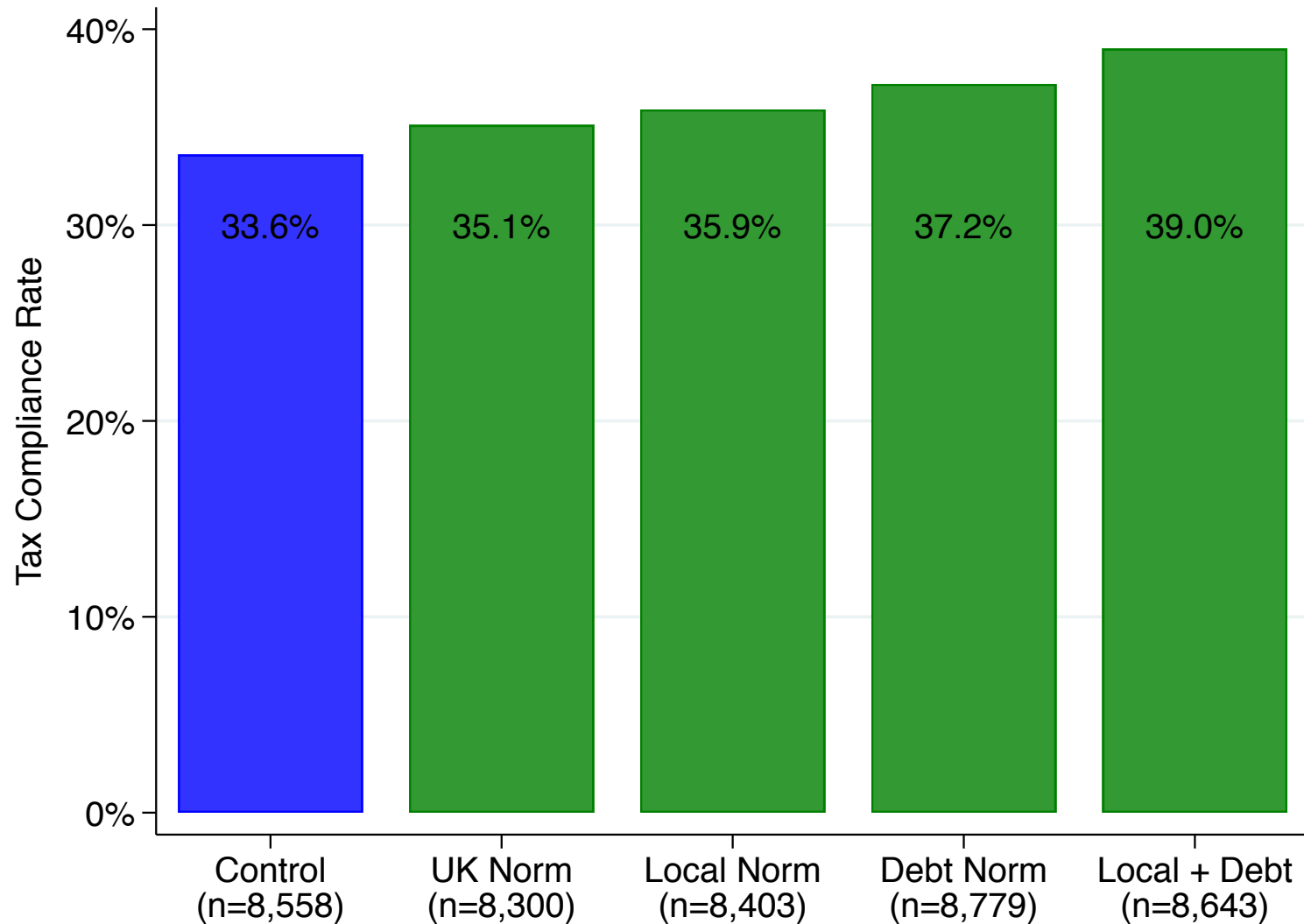
- Tax Compliance

- Most UK taxpayers are in “pay as you earn” system
- But...self-employment, multiple sources of income, trustee status, etc. mean that you have to self-report (about 10 million)
- Those with self-reported income have lowest levels of tax compliance (Kleven et al., 2011)

BIT: Tax Compliance

- 0) Control
- 1) UK Norm: “nine out of ten people in the UK pay their tax on time”
- 2) Local Norm: UK Norm + “most people in your local area pay on time”
- 3) Debt Norm: “most people with a debt like yours have paid it by now”
- 4) Local Norm + Debt Norm

BIT: Tax Compliance Results



BIT: Organ Donation

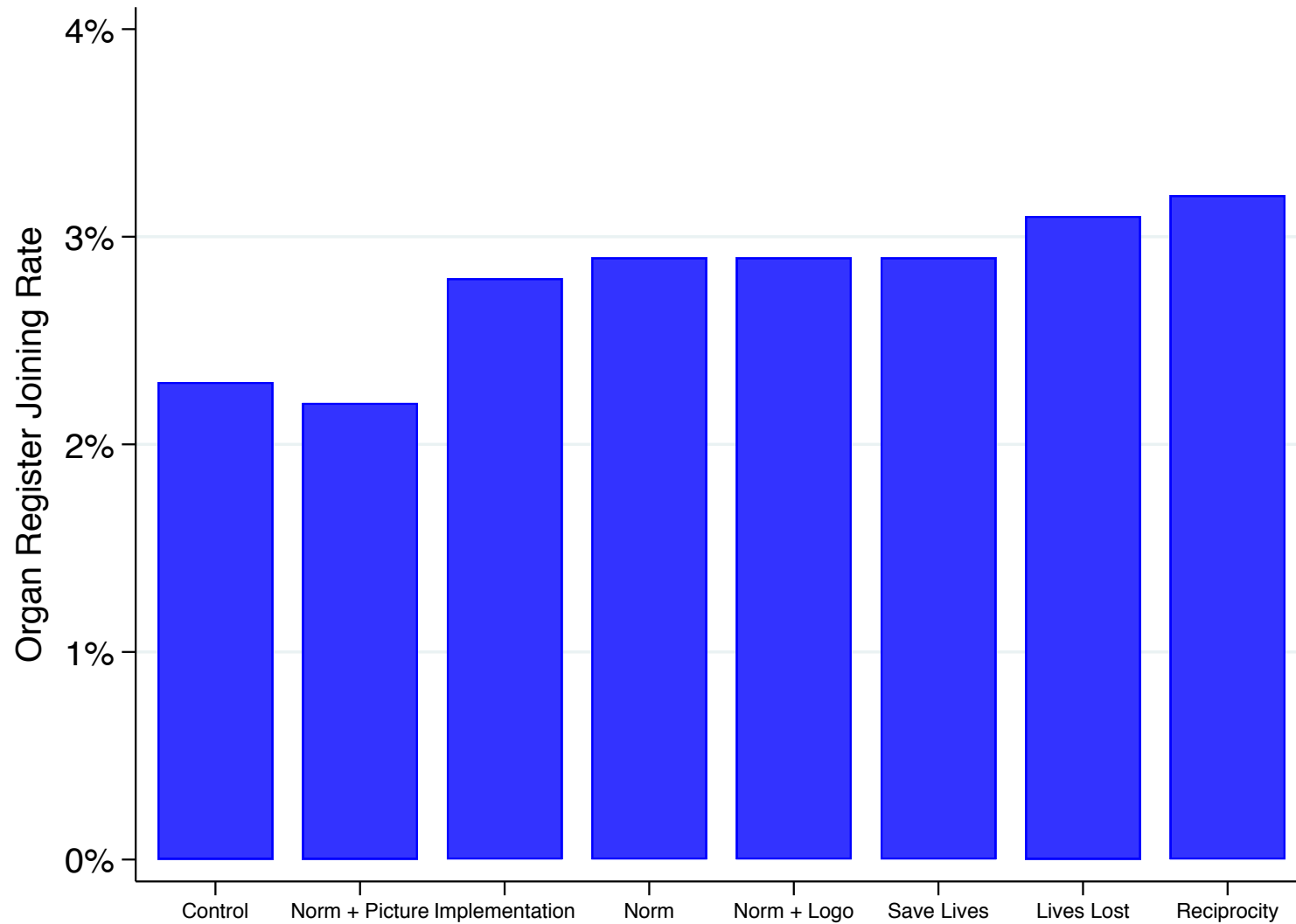
- UK drivers must renew their car tax every year (n=17 million)
- Last page of online process: organ registration



BIT: Organ Donation

- 0) Control: “Thank you. Please join the organ donor register”
- 1) Social: “every day, thousands of people who see this page decide to register”
- 2) Social + picture: picture of a group of happy, healthy people such as might be joining the register
- 3) Norm + logo: brand of the organ donor register
- 4) Loss: benefit of organ donation in terms of loss
- 5) Gain: benefit of organ donation in terms of gain;
- 6) Reciprocity: “If you needed an organ transplant, would you have one? If so, please help others.”
- 7) Implementation-intention gap: “most people say in surveys that they intend to join register to become donors, but just haven’t got round to it.”

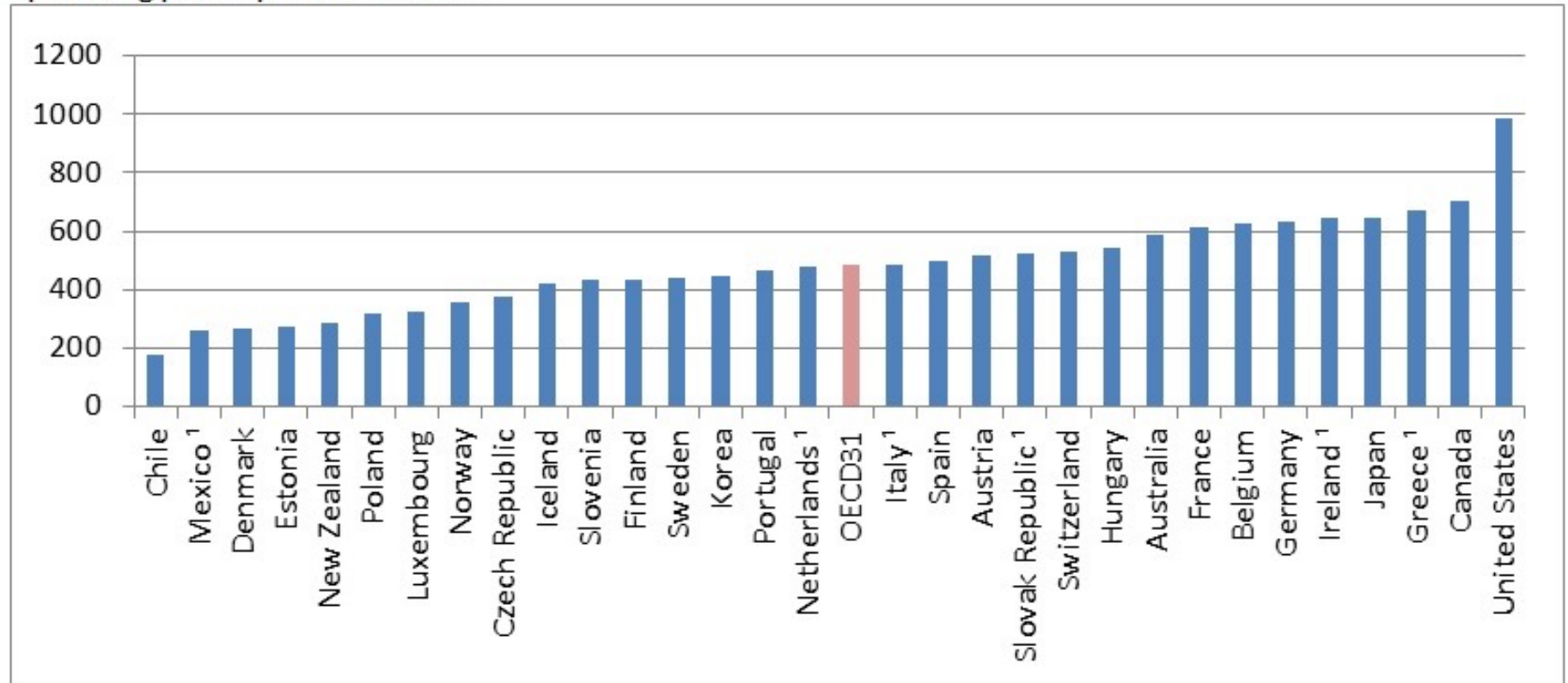
BIT: Organ Donation



Express Scripts: Background

The US spends more on drugs

Spending per capita in PPPUS\$:



Source: [OECD Health Statistics 2013](#)

Express Scripts: Prescription

Pharmacy (30-day fill)



Home delivery (90-day fill)



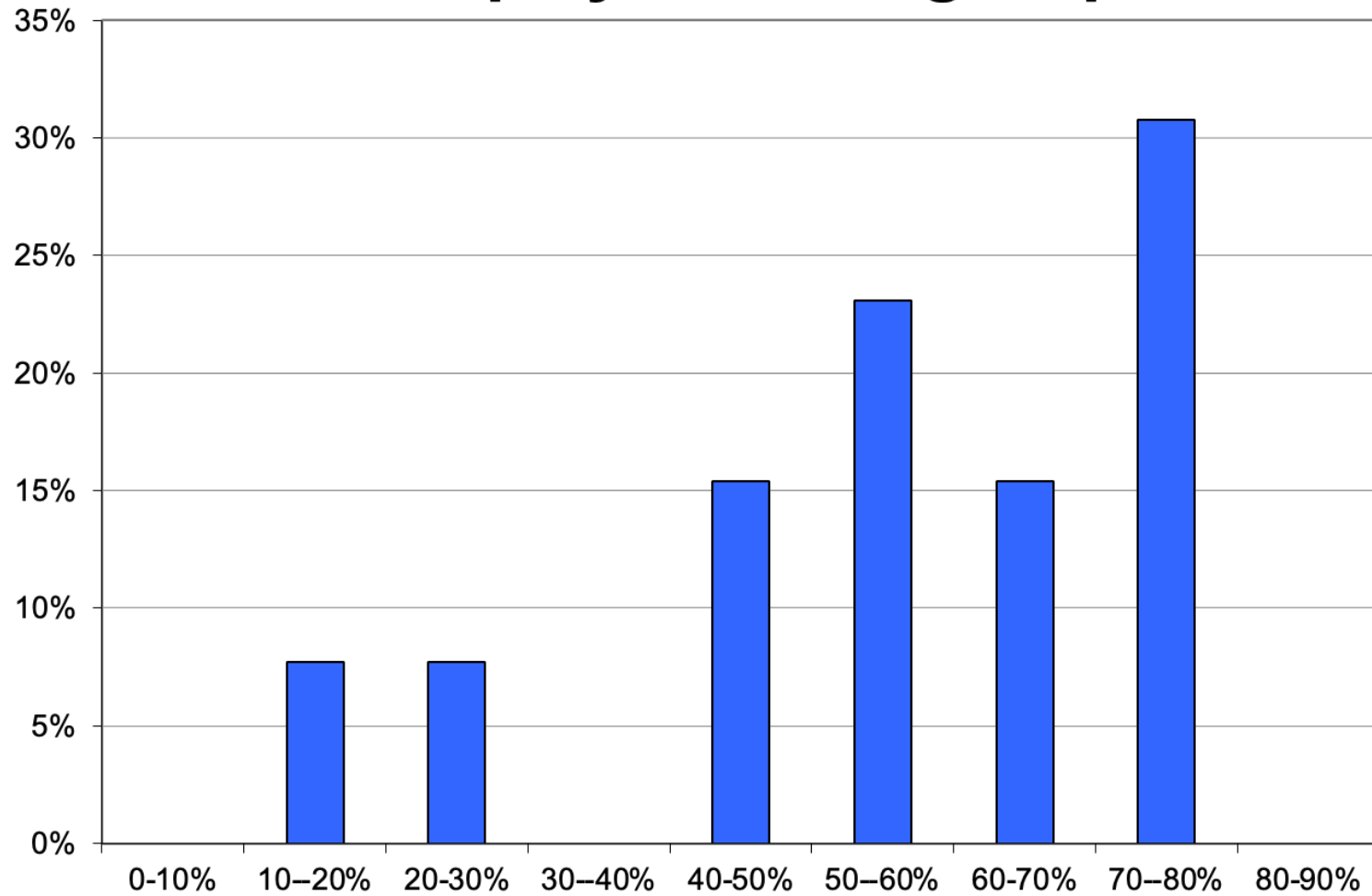
Express Scripts: Copays

Type of Prescription	Pharmacy Pickup (30-day fill)	Home Delivery (90-day fill)
Generics	\$0.27/day	\$0.22/day
Preferred brand (no generic available)	\$0.83/day	\$0.61/day

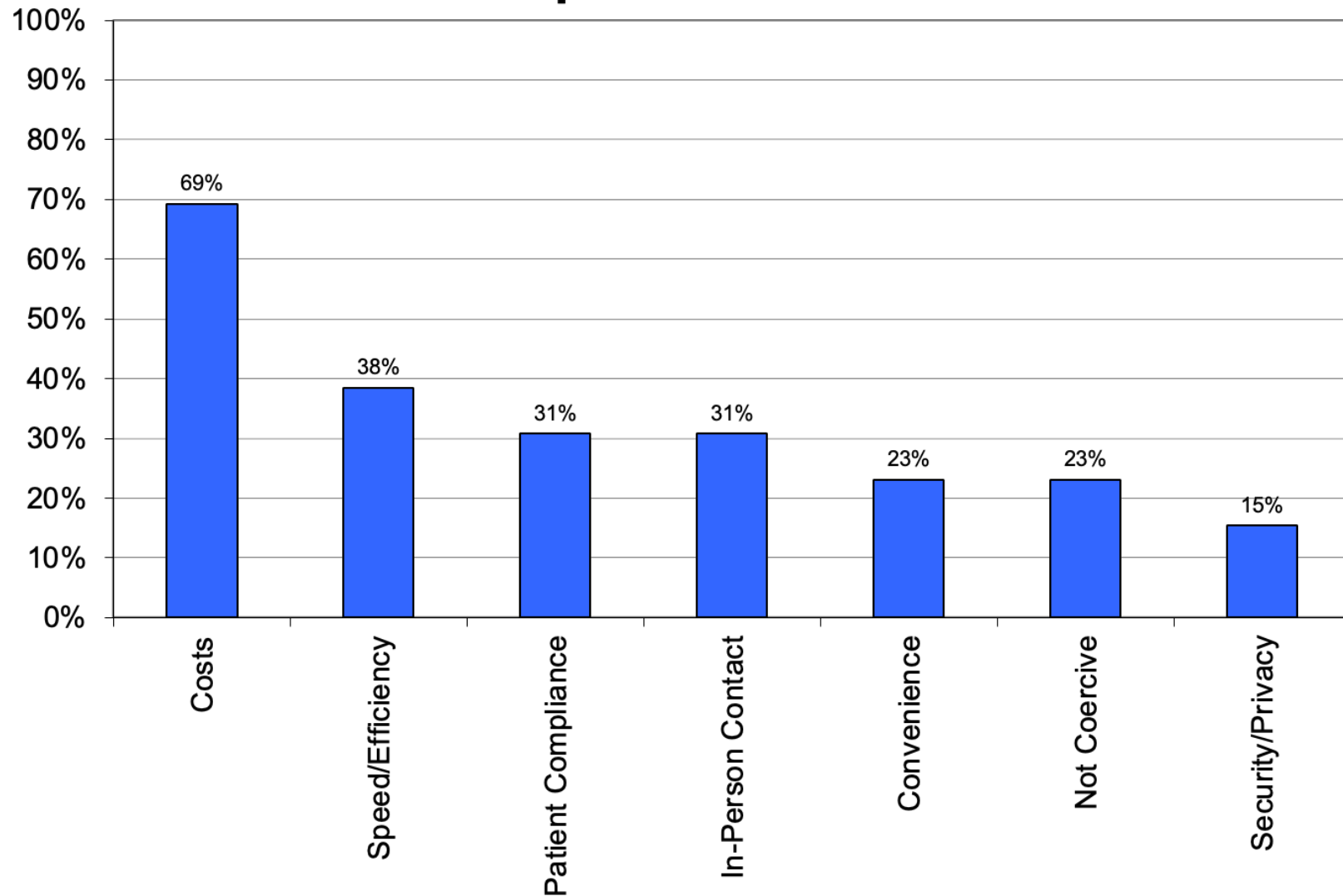
Express Scripts

- The typical pharmacy benefit plan has only 6% of prescriptions filled by home delivery.
- What do you recommend to increase take-up of the home delivery program?

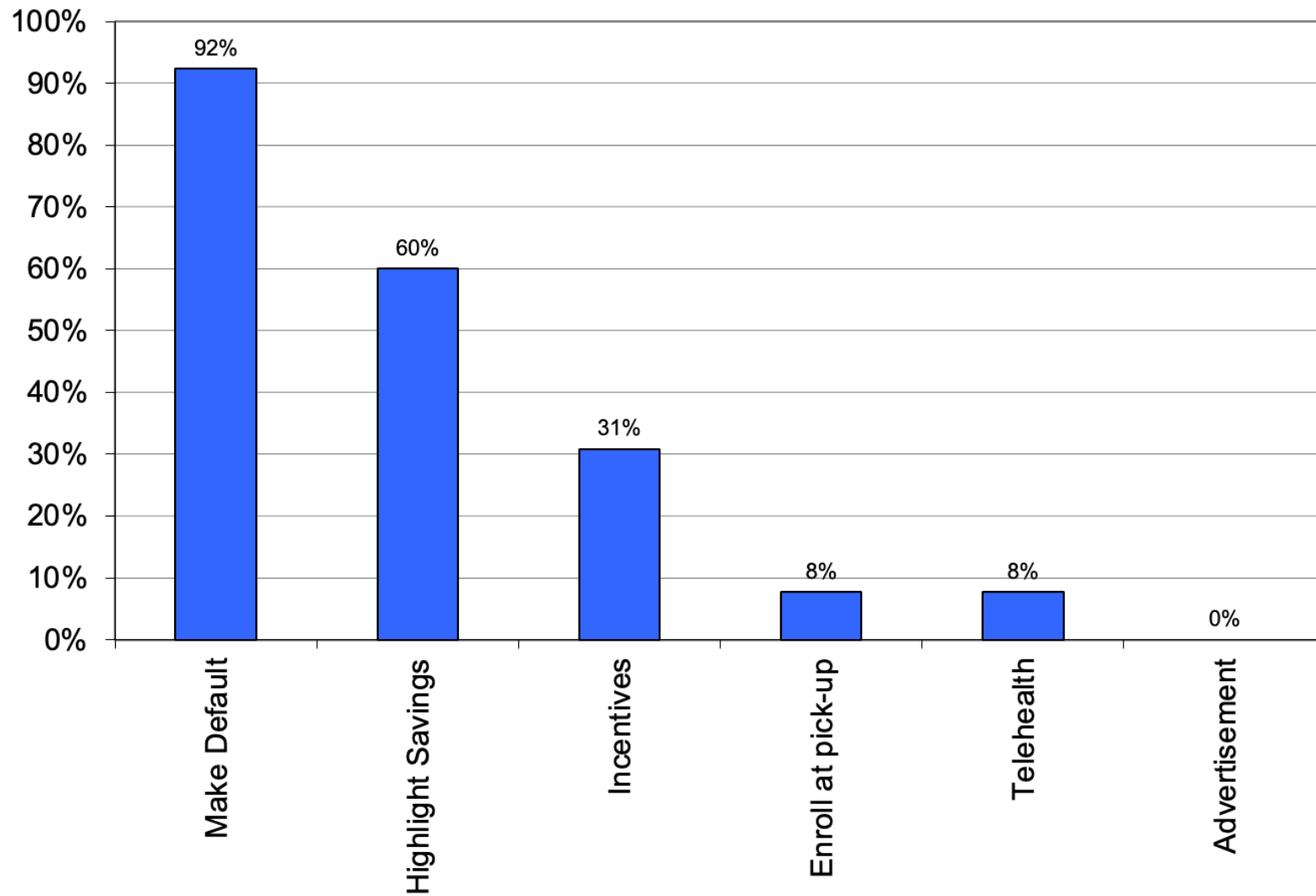
% Employees Taking it up



Most Important Considerations



Recommendations for Bob



Present copayments side-by-side on a 90-day equivalent basis for both retail and home delivery across all tiers; keep underlying copays unchanged. Retail cost currently shows the cost for 30 days, which may make it appear cheaper than the home delivery copayment, even though people may be aware that the figures are not apples-to-apples. (People may just be lazy to run the x3 calculation to confirm which is cheaper.) This would involve almost zero cost, and can be instantly tested.

We recommend introducing an active choice architecture:

- When a maintenance drug is due for refill, patients must choose between home delivery or retail pickup.*
- If no choice is made, the default becomes home delivery for the next fill, but patients can override this with one click.*
- This approach preserves freedom of choice while gently nudging users toward home delivery.*

Using choice architecture and behavioural nudges could incentivise change:

- Making home delivery the default option unless they opt out*
- Highlight the fact that the home delivery guarantees a 90-day worth of drug amount, equivalent to 3 retails visit but at 20% less per month*
- Segmenting employees who are on maintenance drugs, who might have better fit to the programme*
- Create app and notification system to create a seamless experience for home delivery*

What did Express Scripts do?

“Active Choice”

- Identify employees who had a retail pickup for a maintenance medication and did not have a home delivery prescription for that medication on file
 - Send letters and place automated telephone calls to eligible employees in November and December 2008
 - You need to make an **active choice** between retail pickup and home delivery (by April 2009).
-

Active Choice Rationale

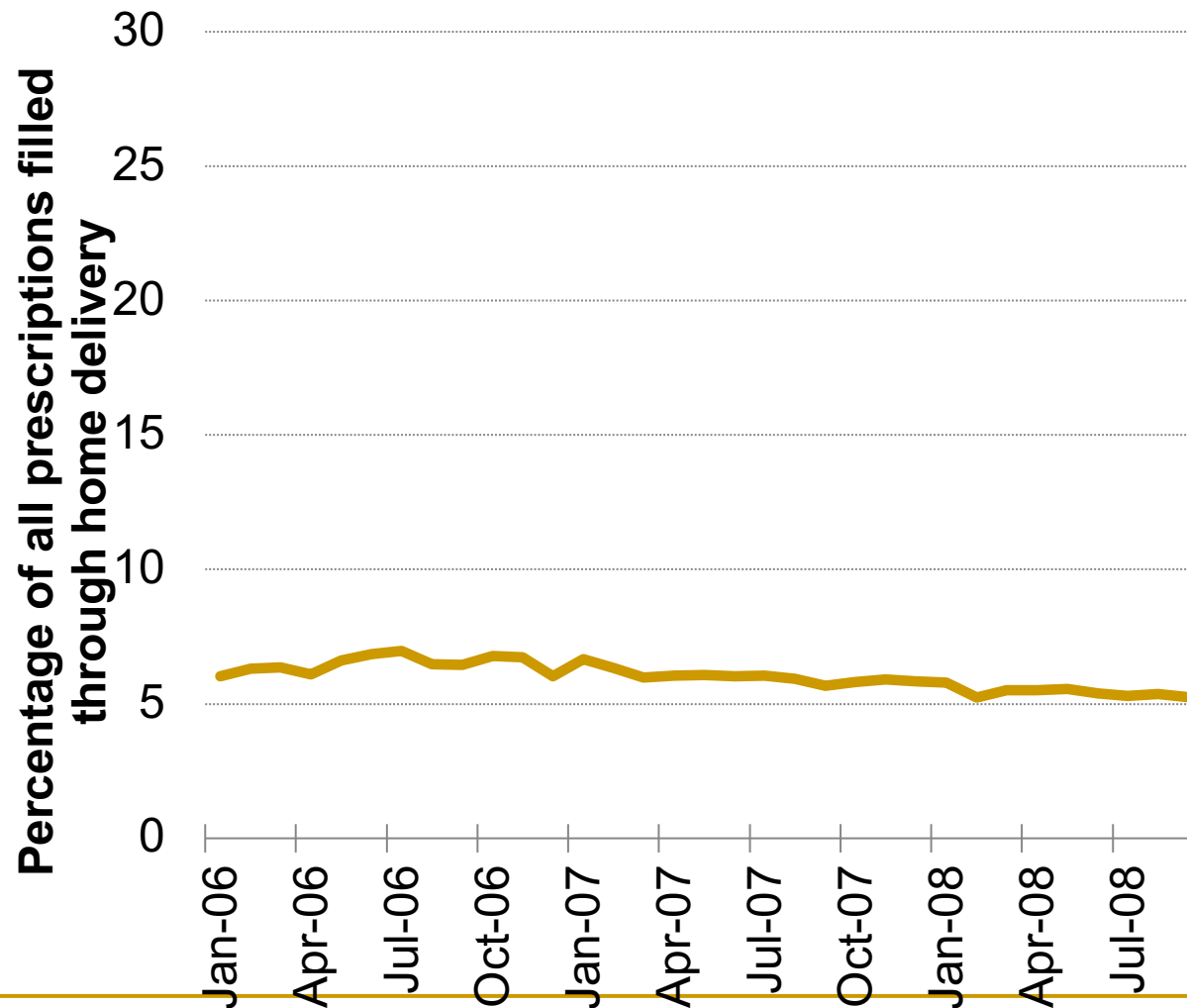
- Preservation of choice and personal agency
 - ❑ Does not default everyone into the same solution
 - ❑ Employees can elect retail pickup or home delivery
 - ❑ Each employee may know what is best for them

Active Choice Prediction

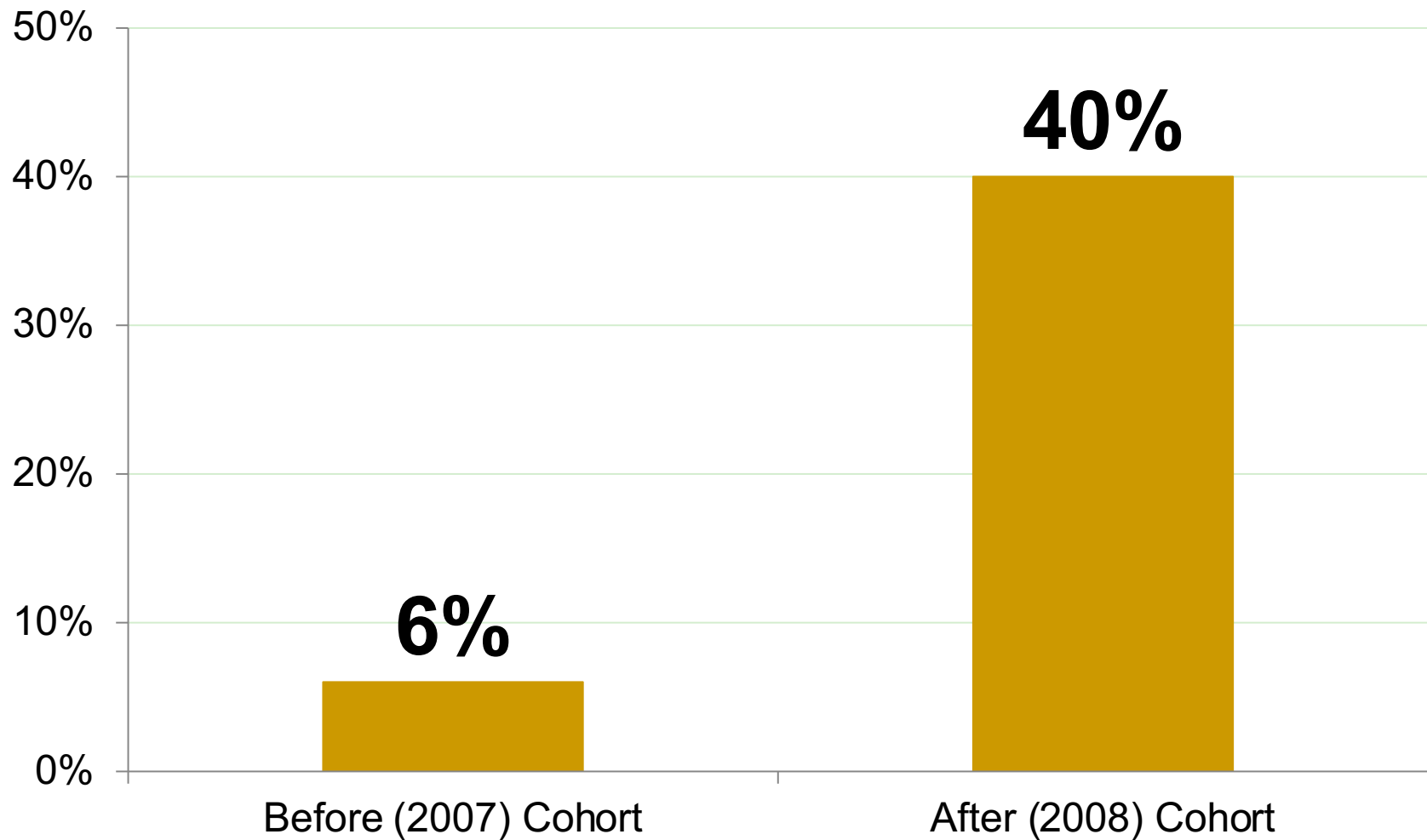
- Percentage (0 to 100) of employees who have at least some home delivery
 - Was formerly 6%
 - .05,.50,.95 fractile



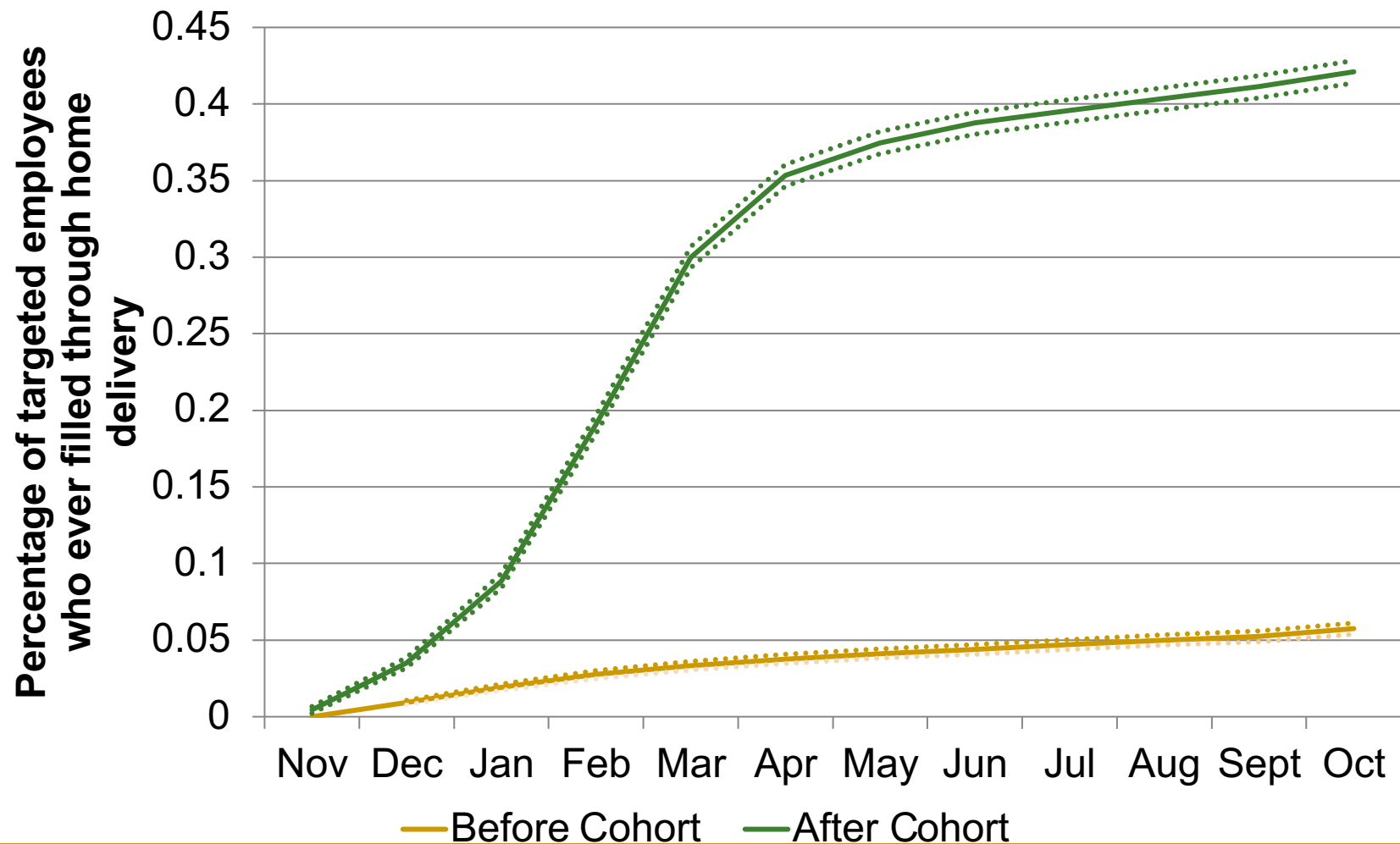
Home Delivery (% of Prescriptions)



Home Delivery (% Employees)



Percentage of Targeted Employees Actively Choosing Home Delivery



Choice Architecture Principles

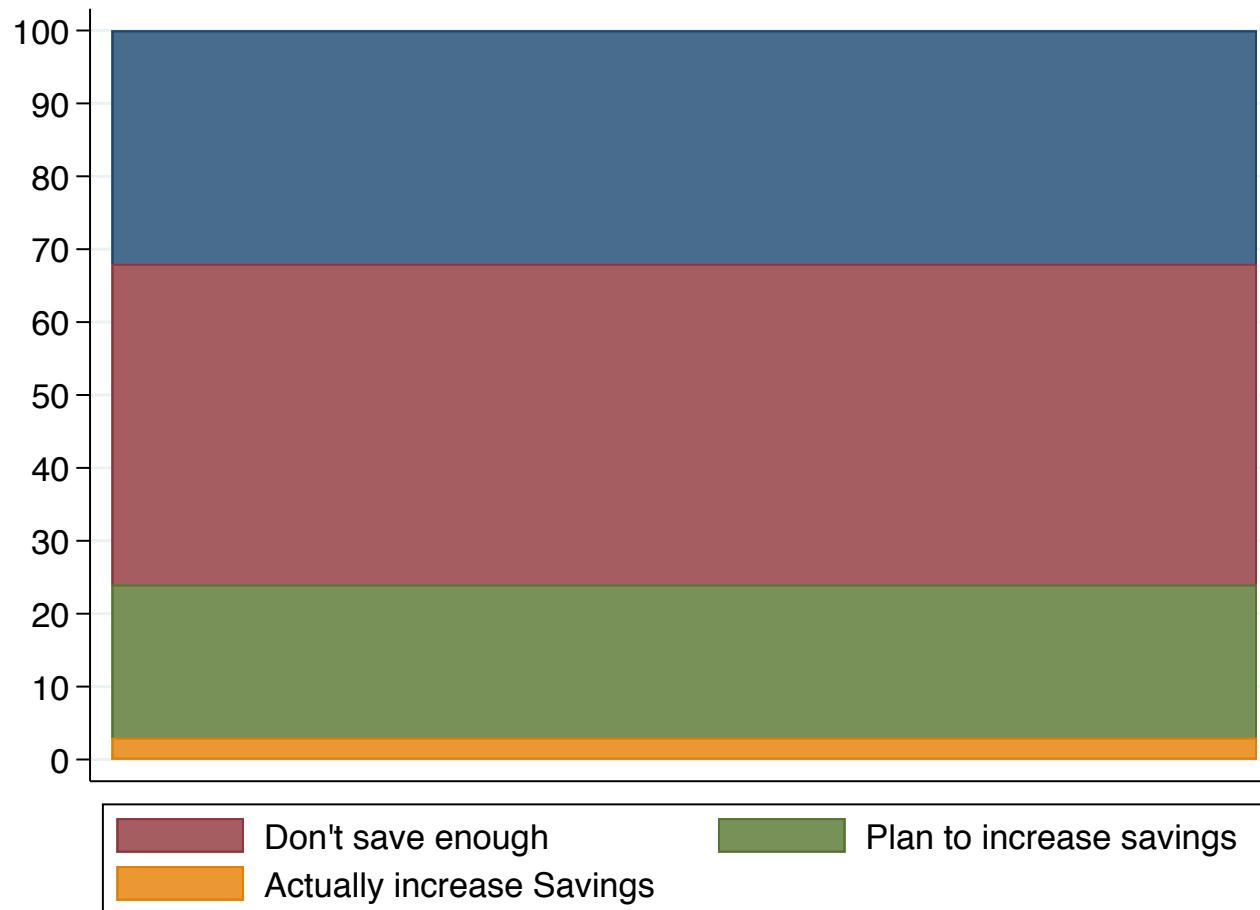
1. Financial incentives may not be cost effective
 2. Information may not be effective
 3. Attention is a scarce resource
 4. Identify the psychological barriers to behavior change
 5. Identify gaps between intentions and actions
 6. Automate when human judgment is *not* needed
 7. Simplify when human judgment is needed
 8. Impose deadlines (and active choices) to fight inertia
 9. Utilize existing contact points and decision triggers
 10. Consider ethical dimensions and societal well-being
-

Five Choice Architecture Levers

- Defaults
- Reminders
- Pre-commitment
- Simplifying choices
- Social norms

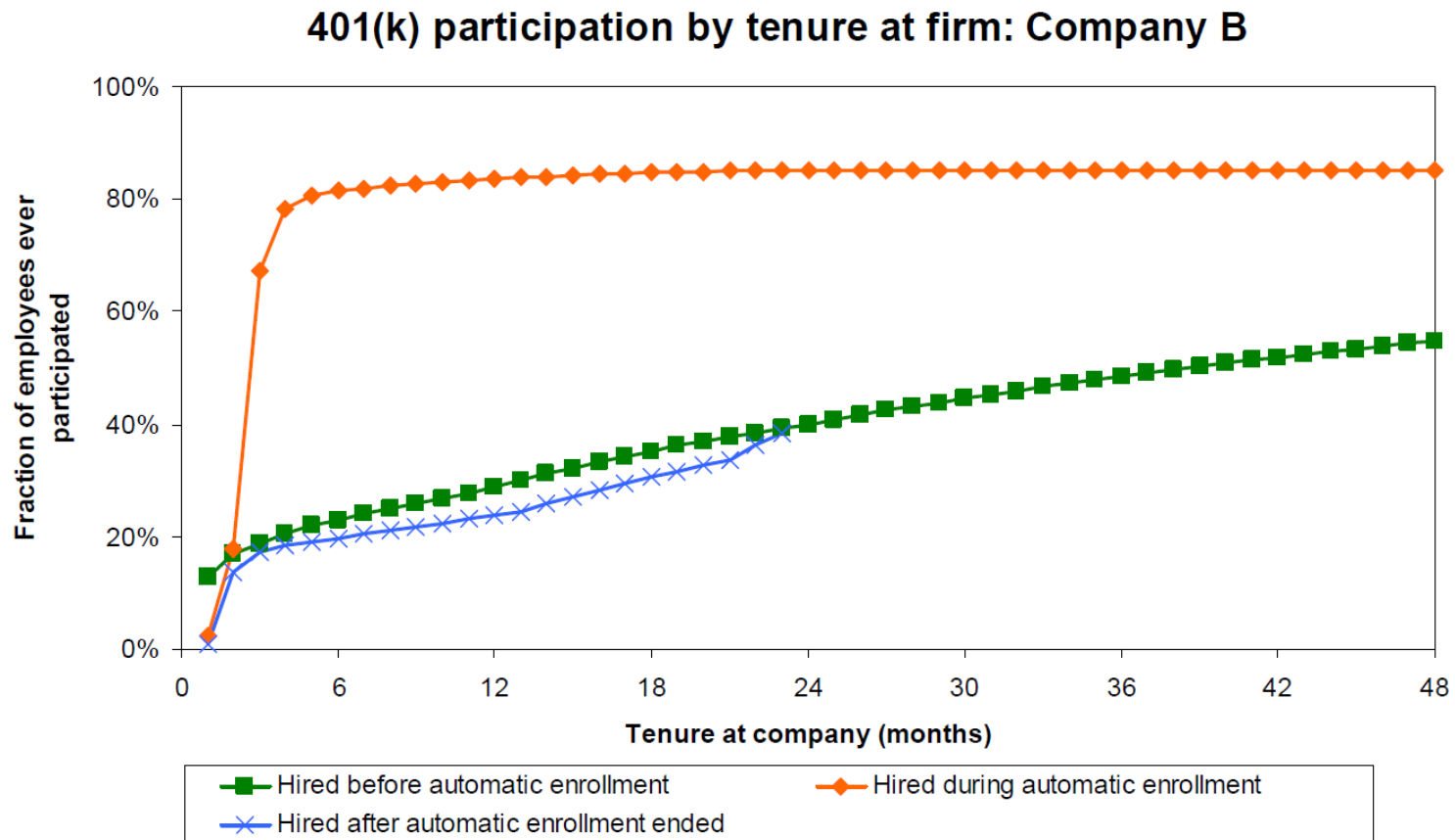
Defaults: Increasing 401(k) Participation

- A Typical Survey of 100 employees



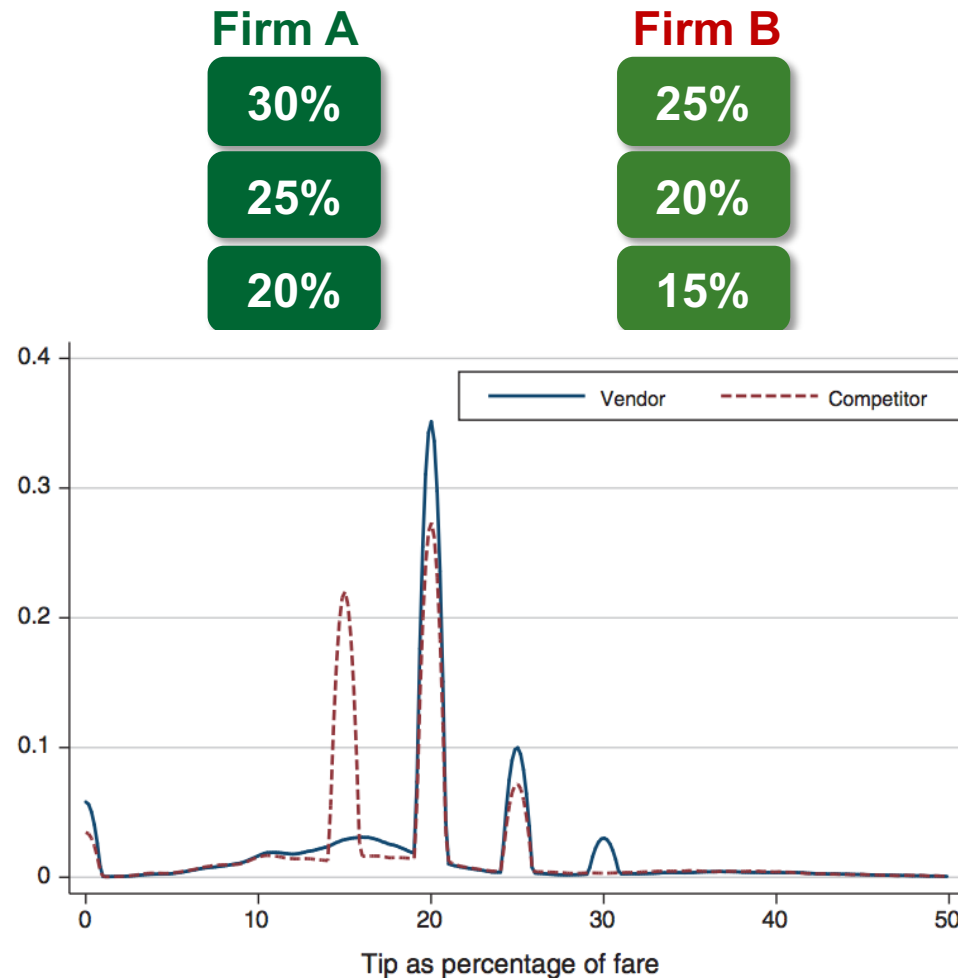
Defaults: Increasing 401(k) Participation

- What if you change the default (Madrian and Shea, 2002)?



Defaults: Changing Options

- Tips and default options in taxis (Haggag and Paci, 2014)



Defaults: Education Performance

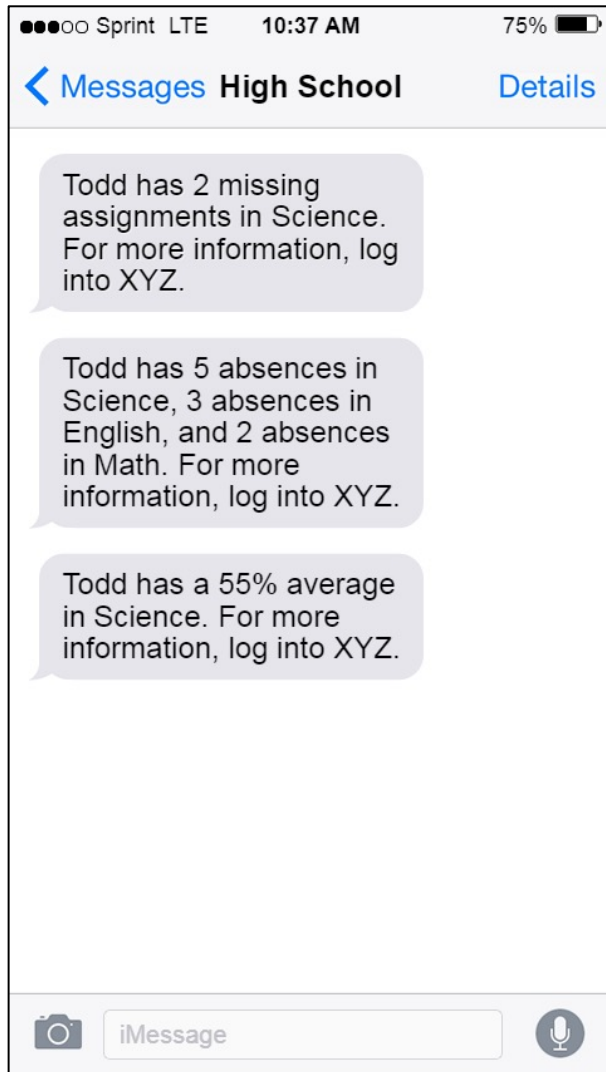
- Changing educational performance through parental involvement (Bergman and Rogers, 2017)

	2012-13		2013-14		2014-15	
	Enrolled	Chronic absences	Enrolled	Chronic absences	Enrolled	Chronic absences
<i>All Students</i>	1,124,520	175,484 (15.6%)	1,140,601	167,837 (14.7%)	1,149,074	183,860 (16.0%)
<i>Low income</i>	553,435	117,225 (21.2%)	569,018	112,823 (19.8%)	570,190	122,051 (21.4%)
<i>Non-low income</i>	589,682	61,313 (10.4%)	591,432	57,854 (9.8%)	597,015	64,550 (10.8%)

Source: Office of the Superintendent for Public Instruction

- How about a text notification system?

Defaults: Education Performance



[School Name] is testing a service that texts you if your child has a low grade, missed assignment or missed class.

Standard

You may enroll by logging in at xyz.com; call school for account info.

Simplified

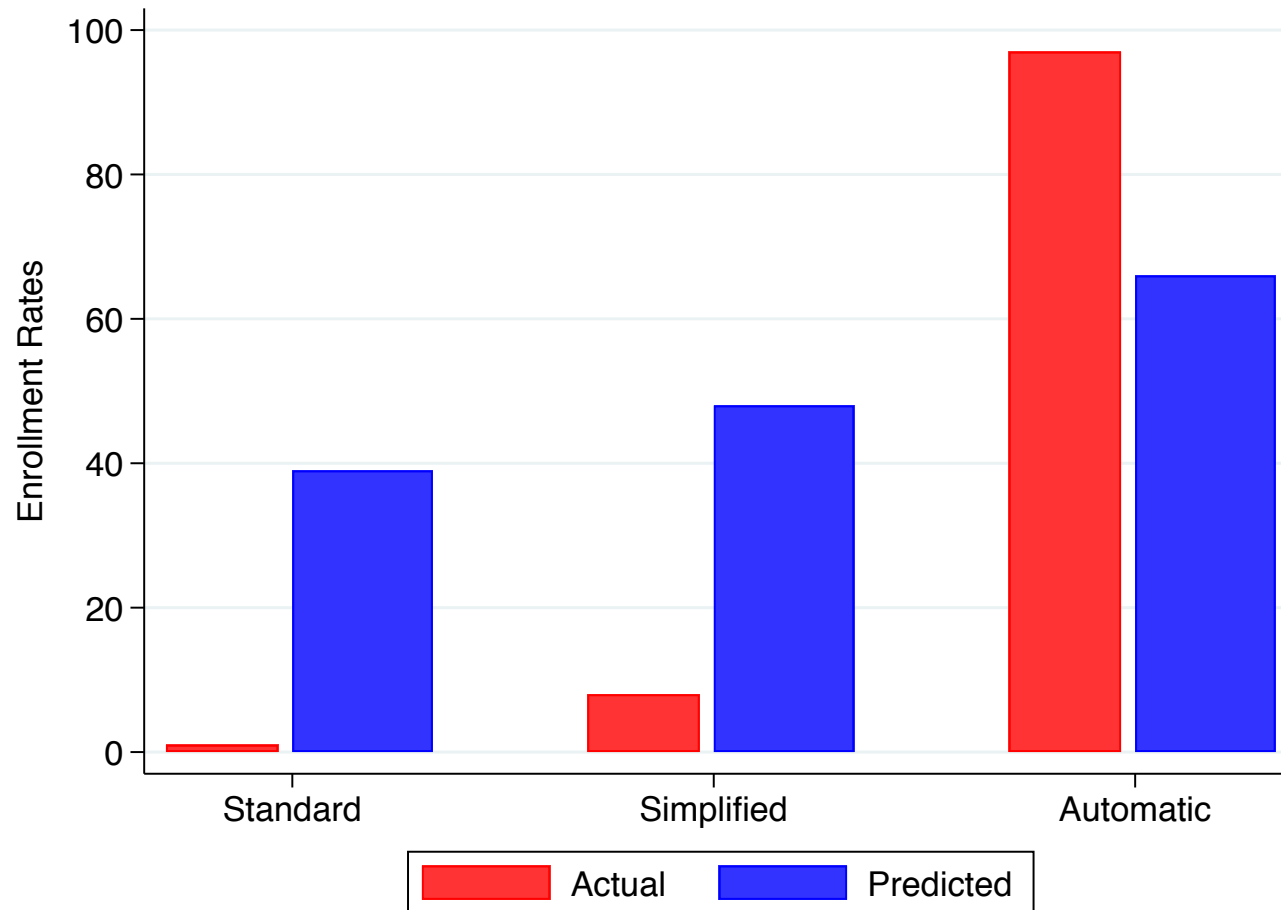
You may enroll by texting "Start" to opt in.

Automatic

You are automatically enrolled: text "Stop" to opt out.

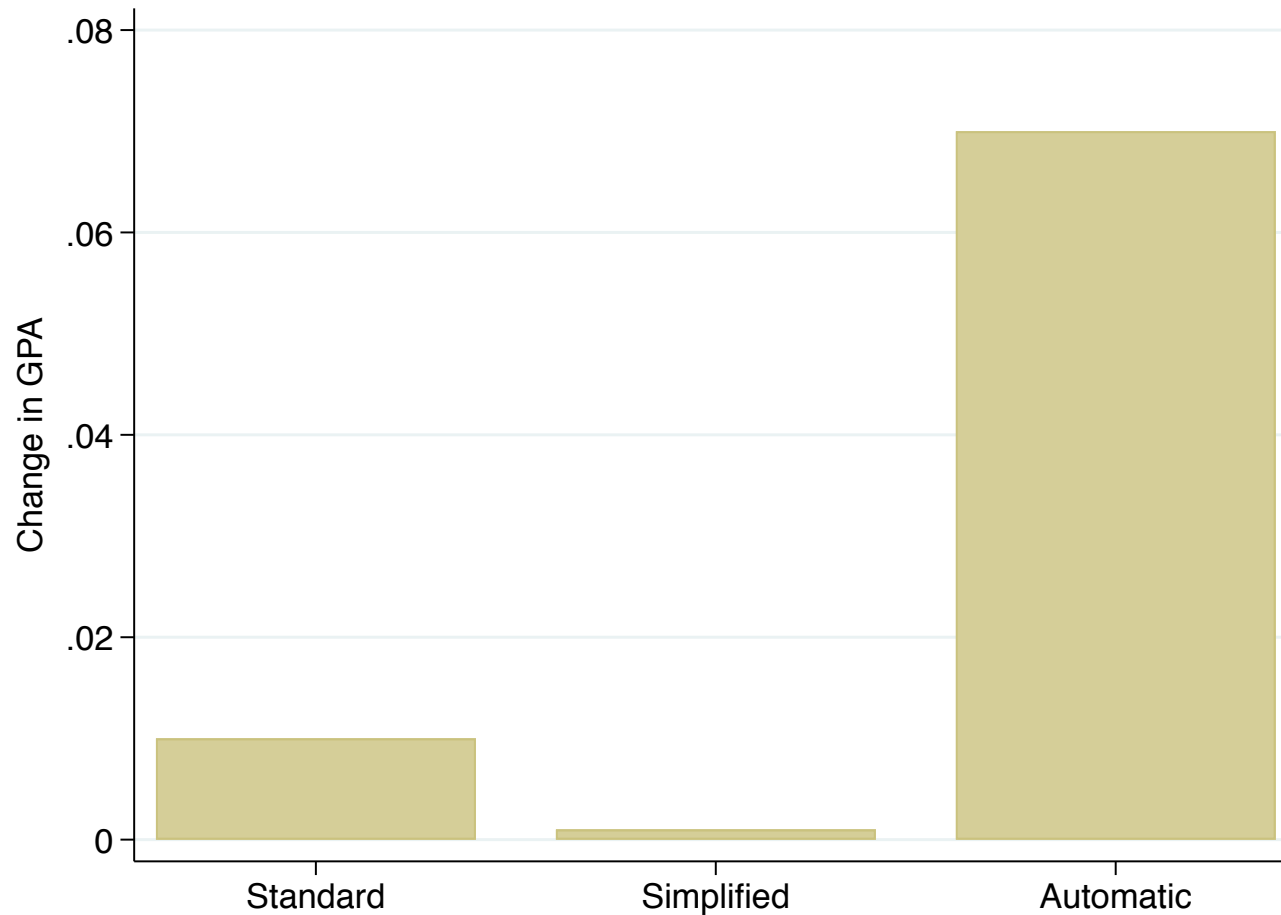
Defaults: Education Performance

- Defaults change participation



Defaults: Education Performance

- Defaults change performance



Reminders

Any of you ever forget...

- To pay a bill?
- A class assignment?
- A meeting (with someone important)?
- Your anniversary?
- Your partner's birthday?
- Your professor's birthday
 - George Wu: October 25
- Your TA's birthday
 - Nick Owsley: November 15



Solution?: Reminders as a Nudge

Reminders: Vaccines

Source: Milkman et al (2021)

- Goal: Boost adoption of seasonal flu vaccine
- Representative statistics (2019-2020)
 - Children (6 months-17 years): 63.7%
 - Adults (≥ 18 years): 48.4%
- Conditions: 19 text-based reminders ($n = 47,306$) + control



Remember to ask for your flu shot

You have an appt w/ Dr. Smith on 10/01 at 11:00 AM & it's flu season. Remember to ask for your flu shot tomorrow.

Flu shot reserved for you

You have an appt w/ Dr. Smith on 10/01 @ 11:00 AM & it's flu season. A flu vaccine is available for you. Protect yourself & your family's health!

Protect yourself by getting a flu shot

It's flu season. At your appt w/ Dr. Smith on 10/01 @ 11:00 AM, be sure to ask for your flu shot.

If you get it, you'll help protect yourself from the flu and the serious complications it can cause, including hospitalization.

Protect others by getting a flu shot

It's flu season. At your appt w/ Dr. Smith on 10/01 @ 11:00 AM, be sure to ask for your flu shot.

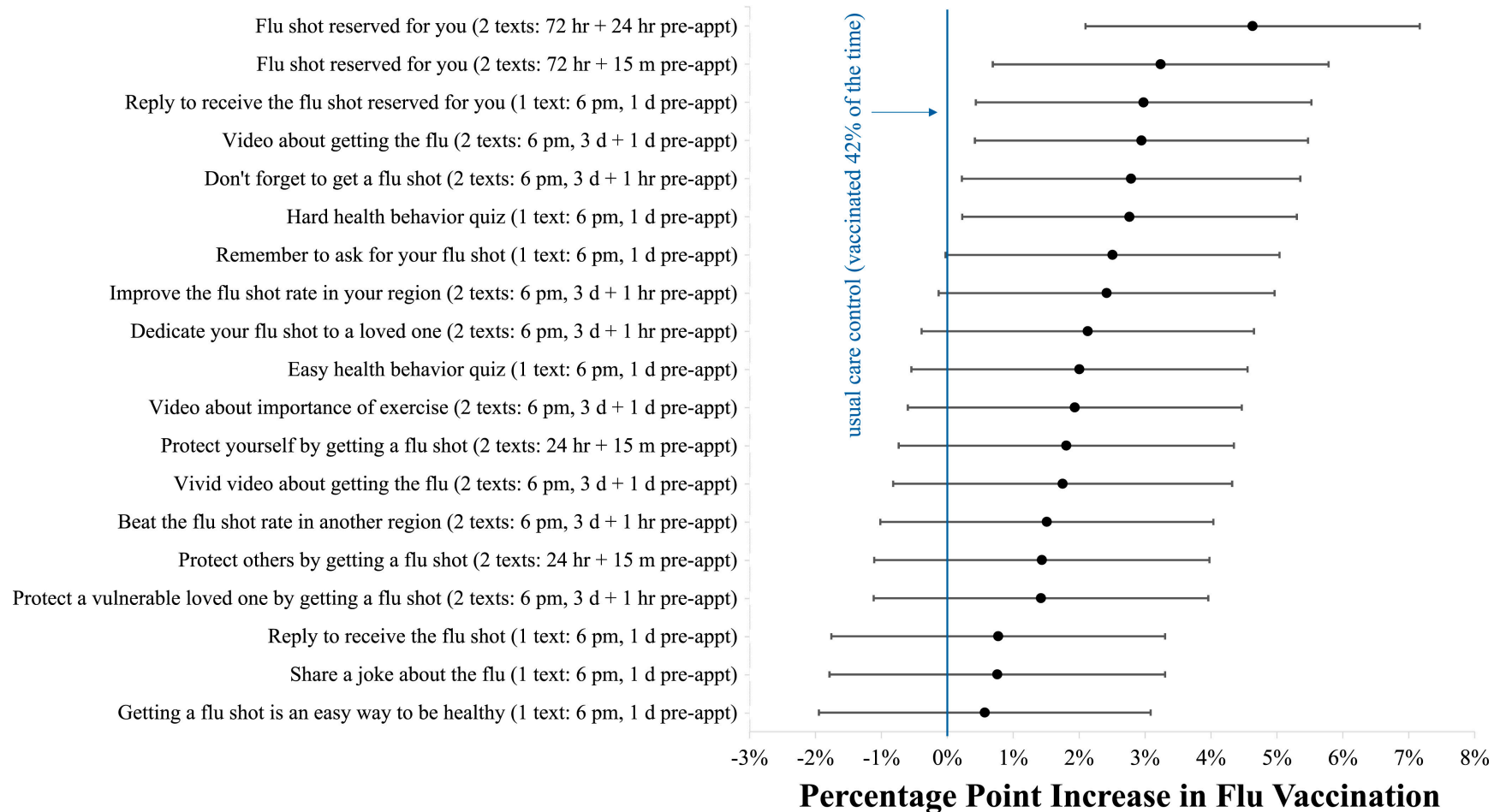
If you get it, you'll help protect your family and friends from the flu and the serious complications it can cause, including hospitalization.

Improve the flu shot rate in your region

In YEAR, flu shots in YOUR REGION lagged behind the target rate of 70%. Help YOUR REGION become a leader in flu prevention and get your flu shot.

Reminders: Vaccines

Source: Milkman et al (2021)



Reminders: COVID

Source: Dai et al (2021)

- Goal: Boost adoption of COVID-19 vaccine
- Details: UCLA, January 29, 2021+
- Conditions: 4 text-based reminders ($n = 93,354$) + control



Ownership + Video

UCLA Health: [participant's first name], a COVID-19 vaccine has just been made available to you at UCLA Health. Please take 2 simple steps: 1. Watch this important 2 min video: [link]. 2. Claim your dose today by making a vaccination appointment here: uclahealth.org/schedule.

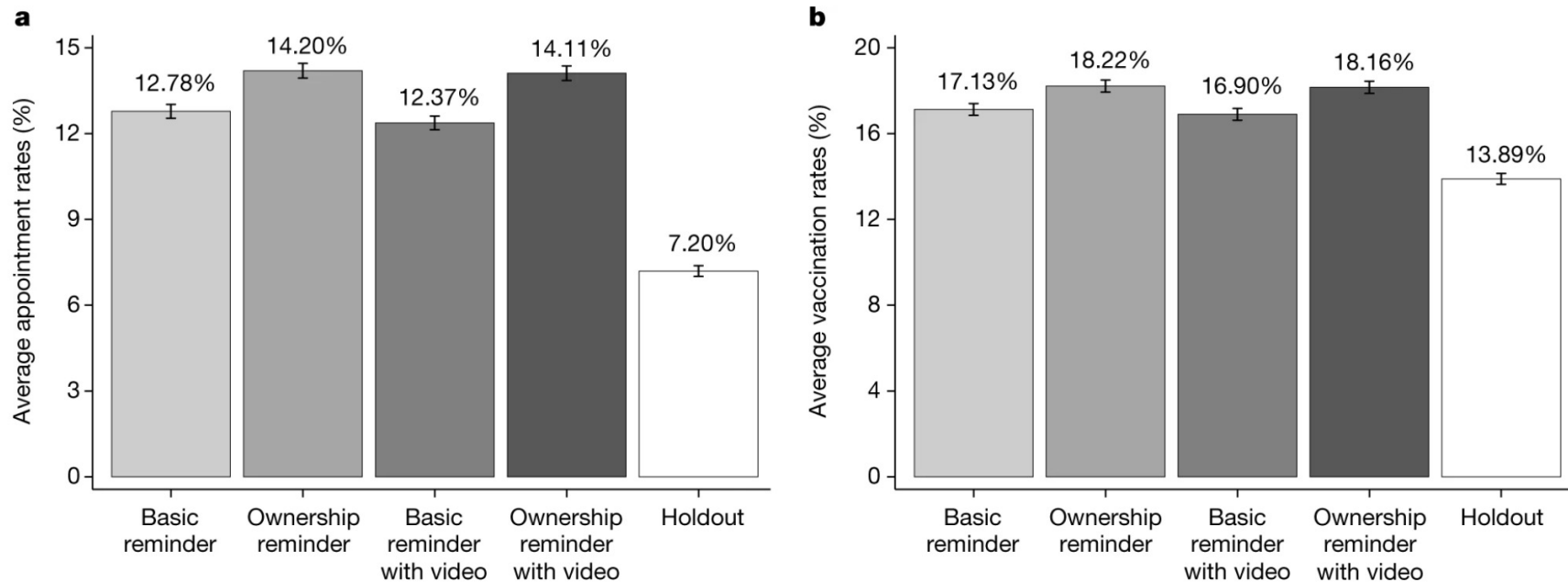
‘UCLA Health: [participant's first name], you can get the COVID-19 vaccine at UCLA Health. Make a vaccination appointment here: uclahealth.org/schedule.’

Basic Reminder

‘UCLA Health: [participant’s first name], you can get the COVID-19 vaccine at UCLA Health. Make a vaccination appointment here: uclahealth.org/schedule.’

Reminders: COVID

Source: Dai et al (2021)



a, b, Proportion of participants in each condition who scheduled an appointment for the first dose of the COVID-19 vaccine at UCLA Health between 15:00 h on the first reminder date and 23:59 h on the fifth day following the first reminder date (**a**) and the proportion of participants in each condition who obtained the first dose of the COVID-19 vaccine at UCLA Health within four weeks of the first reminder date (**b**). Error bars represent ± 1 s.e.m. The number of participants in each condition (from left to right in each panel) is 18,629, 18,592, 18,757, 18,627 and 18,749.

Pre-Commitment

You want to start a diet

- Who wants to start tonight?
- Who wants to start in a week?

Three observations

- It's easier to be virtuous later than now
- When it is later, we usually are not virtuous (because it is now “now”)
- We can make decisions *now* that make it easier to be virtuous *later*

Example: Spending \$\$\$ on a gym membership rather than paying as you go

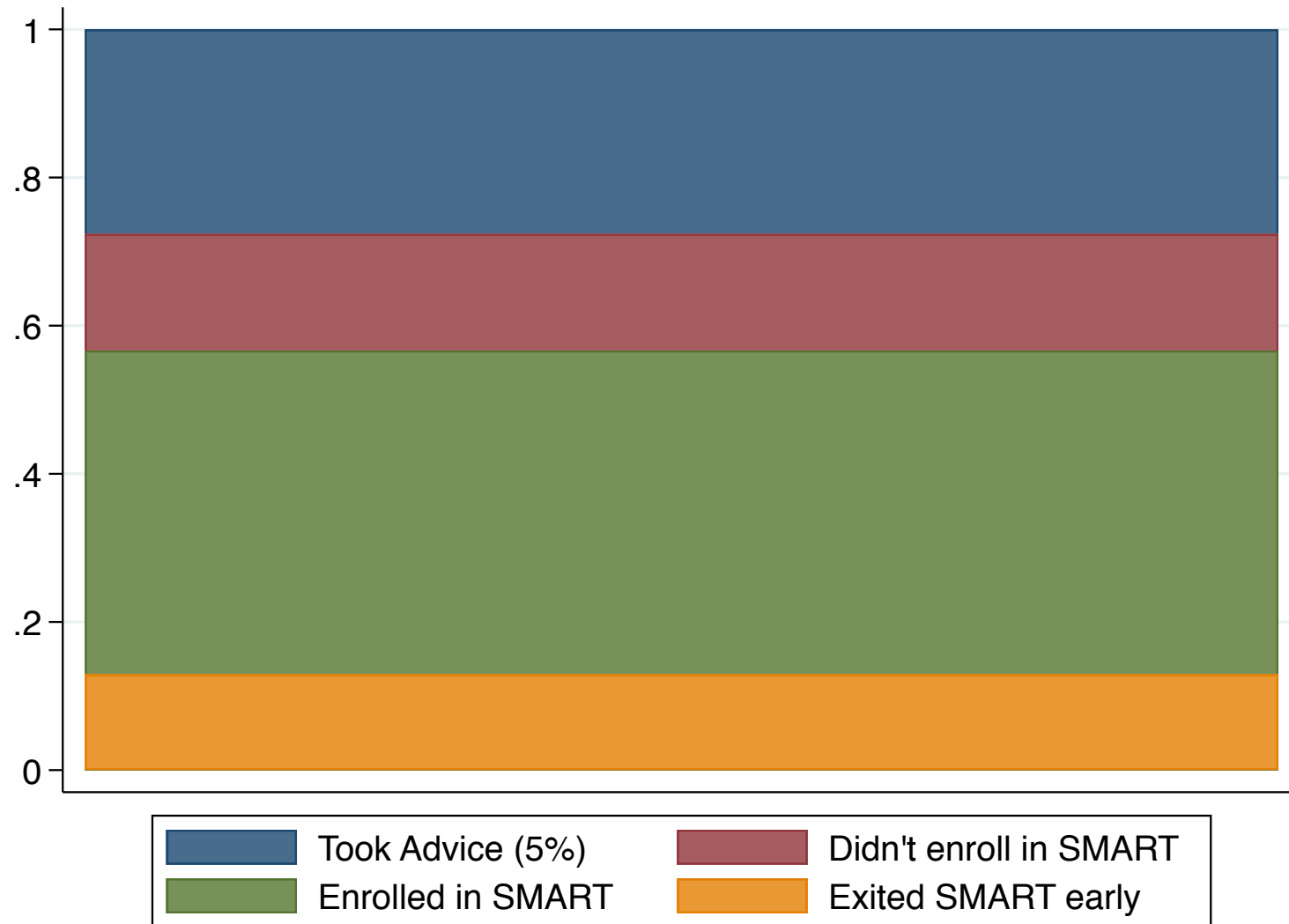
“Save More Tomorrow”

Study: Manufacturing firm (Benartzi and Thaler, 2004)

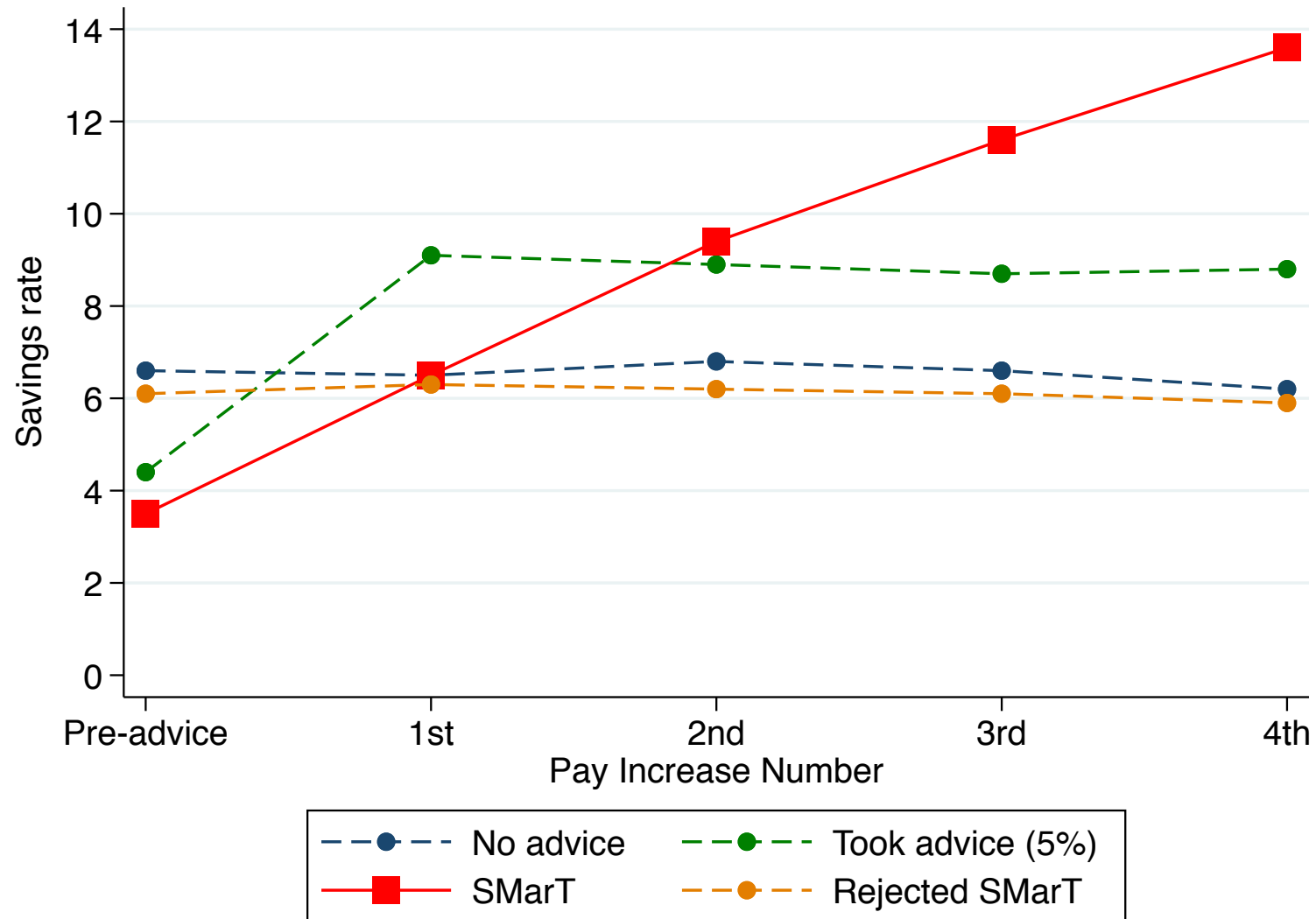
Steps

- Employees meet a financial consultant who suggests a 5% increase in savings
 - Save More Tomorrow (“SMarT”)
 - Participants pre-commit to increase their saving rate by 3% per year
 - Saving increases are synchronized with pay raises
 - The increases continue unless the participant opts out or hits the plan max
-

SMART Results: Participation

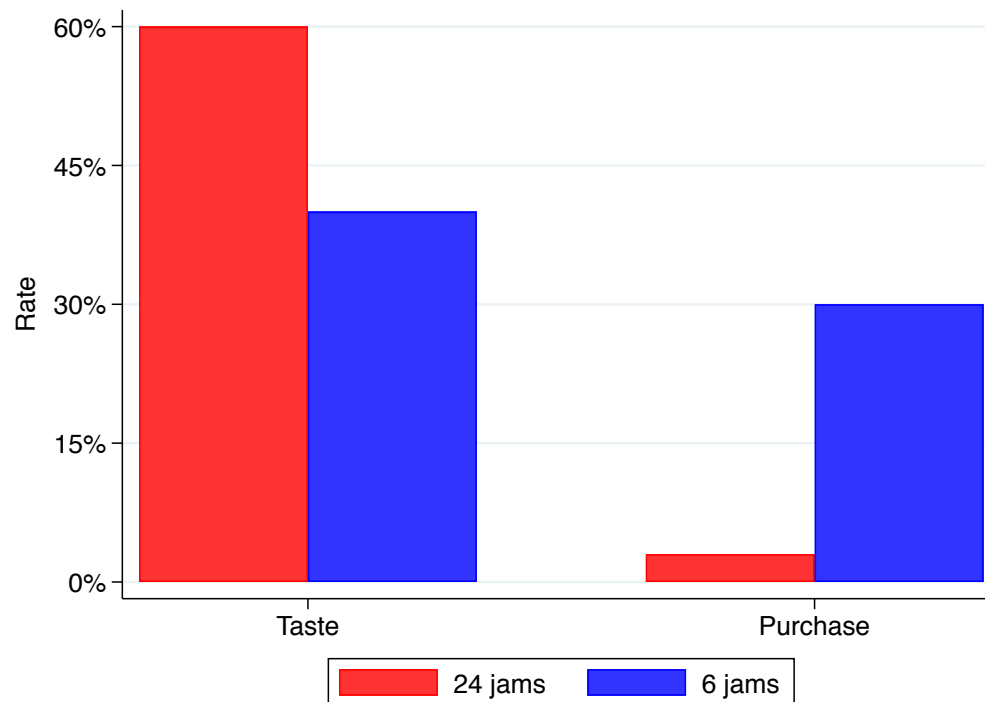


SMarT Results: Savings Rates



Simplifying choice

- Iyengar and Lepper (2000)
- Two tasting booths
 - Many options: 24 jams
 - Few options: 6 jams



Participation Rates in 401(k)

- Vanguard Center for Retirement Research (Iyengar, Huberman, & Jiang, 2004)
 - Data (2001)
 - 793,794 employees
 - 647 defined contribution plans
 - Plan options: 2 to 59 (median is 13)
 - Regression analysis to understand relationship between participation rates and number of plans
 - 2 options – 75%
 - 59 options – 61%

Simplifying Choice

- Study: Examine Health Care Plan Choices (Bhargava, Loewenstein, and Sydnor, 2017)

Build Your Own Medical Option
1. Which annual deductible meets your needs?
(Remember, a lower deductibles mean higher annual paycheck costs.)

	Deductible Individual/Family
☐	In Network \$350 Individual; \$700 Family Out of Network \$3,500 Individual; \$7,000 Family
☐	In Network \$500 Individual; \$1,000 Family Out of Network \$3,500 Individual; \$7,000 Family
☐	In Network \$750 Individual; \$1,500 Family Out of Network \$3,500 Individual; \$7,000 Family
☐	In Network \$1,000 Individual; \$2,000 Family Out of Network \$3,500 Individual; \$7,000 Family

The Menu of 48 Plan Options

Moop/Coins/ Copay	Deductible	Premium	Deductible Diff	Prem Diff 1k	Moop/Coins/ Copay	Deductible	Premium	Deductible Diff	Prem Diff 1k
1.5k/90/15	\$1,000	\$930	--	--	2.5k/80/15	\$1,000	\$713	--	--
1.5k/90/15	\$750	\$1,463	\$250	\$533	2.5k/80/15	\$750	\$1,217	\$250	\$505
1.5k/90/15	\$500	\$1,568	\$500	\$638	2.5k/80/15	\$500	\$1,315	\$500	\$602
1.5k/90/15	\$350	\$2,134	\$650	\$1,204	2.5k/80/15	\$350	\$1,889	\$650	\$1,176
1.5k/90/25	\$1,000	\$761	--	--	2.5k/80/25	\$1,000	\$662	--	--
1.5k/90/25	\$750	\$1,377	\$250	\$616	2.5k/80/25	\$750	\$1,168	\$250	\$506
1.5k/90/25	\$500	\$1,474	\$500	\$713	2.5k/80/25	\$500	\$1,252	\$500	\$590
1.5k/90/25	\$350	\$2,047	\$650	\$1,286	2.5k/80/25	\$350	\$1,808	\$650	\$1,146
1.5k/80/15	\$1,000	\$877	--	--	3k/90/15	\$1,000	\$657	--	--
1.5k/80/15	\$750	\$1,408	\$250	\$531	3k/90/15	\$750	\$1,260	\$250	\$603
1.5k/80/15	\$500	\$1,497	\$500	\$620	3k/90/15	\$500	\$1,378	\$500	\$722
1.5k/80/15	\$350	\$2,037	\$650	\$1,160	3k/90/15	\$350	\$1,912	\$650	\$1,255
1.5k/80/25	\$1,000	\$817	--	--	3k/90/25	\$1,000	\$654	--	--
1.5k/80/25	\$750	\$1,321	\$250	\$504	3k/90/25	\$750	\$1,173	\$250	\$519
1.5k/80/25	\$500	\$1,419	\$500	\$602	3k/90/25	\$500	\$1,252	\$500	\$598
1.5k/80/25	\$350	\$1,957	\$650	\$1,140	3k/90/25	\$350	\$1,778	\$650	\$1,124
2.5k/90/15	\$1,000	\$748	--	--	3k/80/15	\$1,000	\$641	--	--
2.5k/90/15	\$750	\$1,333	\$250	\$585	3k/80/15	\$750	\$1,089	\$250	\$448
2.5k/90/15	\$500	\$1,455	\$500	\$707	3k/80/15	\$500	\$1,152	\$500	\$511
2.5k/90/15	\$350	\$1,983	\$650	\$1,235	3k/80/15	\$350	\$1,608	\$650	\$967
2.5k/90/25	\$1,000	\$656	--	--	3k/80/25	\$1,000	\$634	--	--
2.5k/90/25	\$750	\$1,235	\$250	\$579	3k/80/25	\$750	\$1,038	\$250	\$404
2.5k/90/25	\$500	\$1,372	\$500	\$716	3k/80/25	\$500	\$1,114	\$500	\$480
2.5k/90/25	\$350	\$1,862	\$650	\$1,206	3k/80/25	\$350	\$1,605	\$650	\$971

Shaded options are dominated by some non-shaded option!

How did people choose?

- A **majority** of the 23,894 employees selected financially dominated plans (even though default plan was a non-dominated plan!)
 - The dominated plans are low deductible (LD)
 - 13% of employees with \$750 deductible: Paying \$528 to avoid a maximum of \$250 in out-of-pocket costs.
 - On average, switching to a \$1000 deductible plan saves \$372



What would you do to get employees to choose high deductible plans?

Social Norms

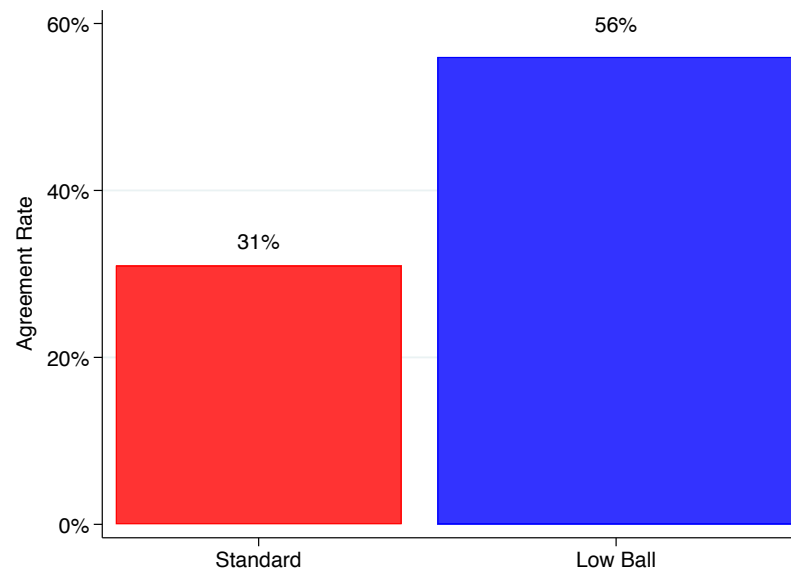
Much of this reviews Managerial Psychology (Class 5?, Persuasion and Influence)

- Commitment and Consistency
- Social Proof
- Similarity and Liking

Commitment and Consistency

Study: Recruitment of students to participate in 7:00 AM Study (Cialdini, Cacioppo, Bassett & Miller, 1978)

- (Control) “Would you like to come to the lab at 7 AM...?”
- (Low Ball) “Would you like to participate in a psychology study...”? “The Study is at 7 AM.”



Commitment and Consistency

Study: Red Cross Donations (n=19,426) (Kessler and Milkman, 2013)

- Remind people of past donations

Reminder of *Your* Past Behavior

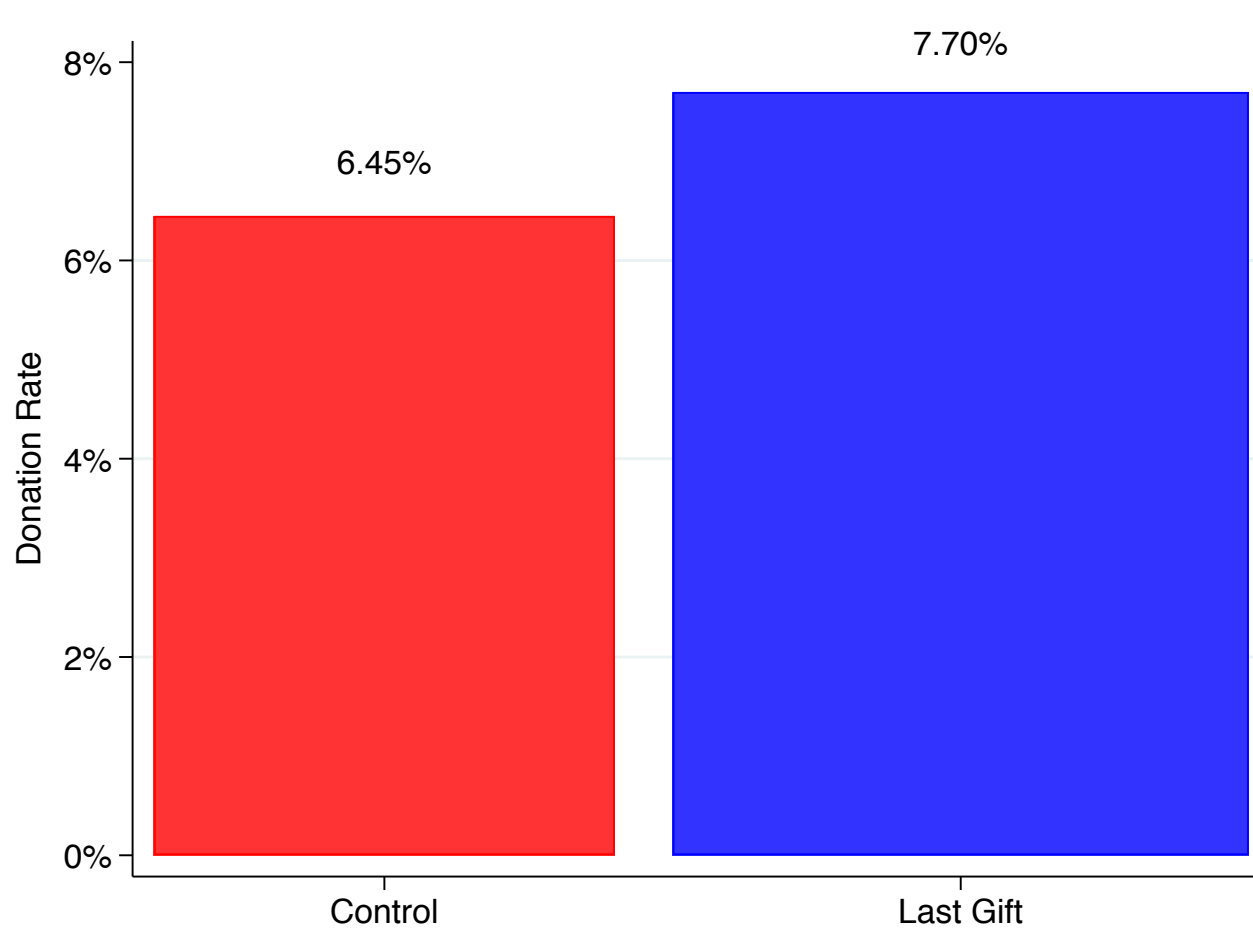


Control



Commitment and Consistency

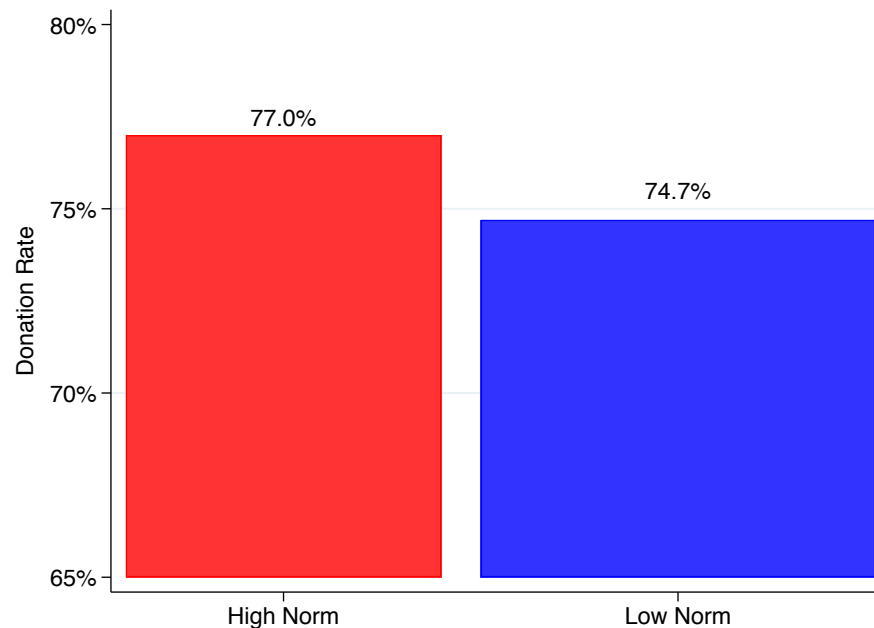
Study: Red Cross Donations (n=19,426) (Kessler and Milkman, 2013)



Social Proof

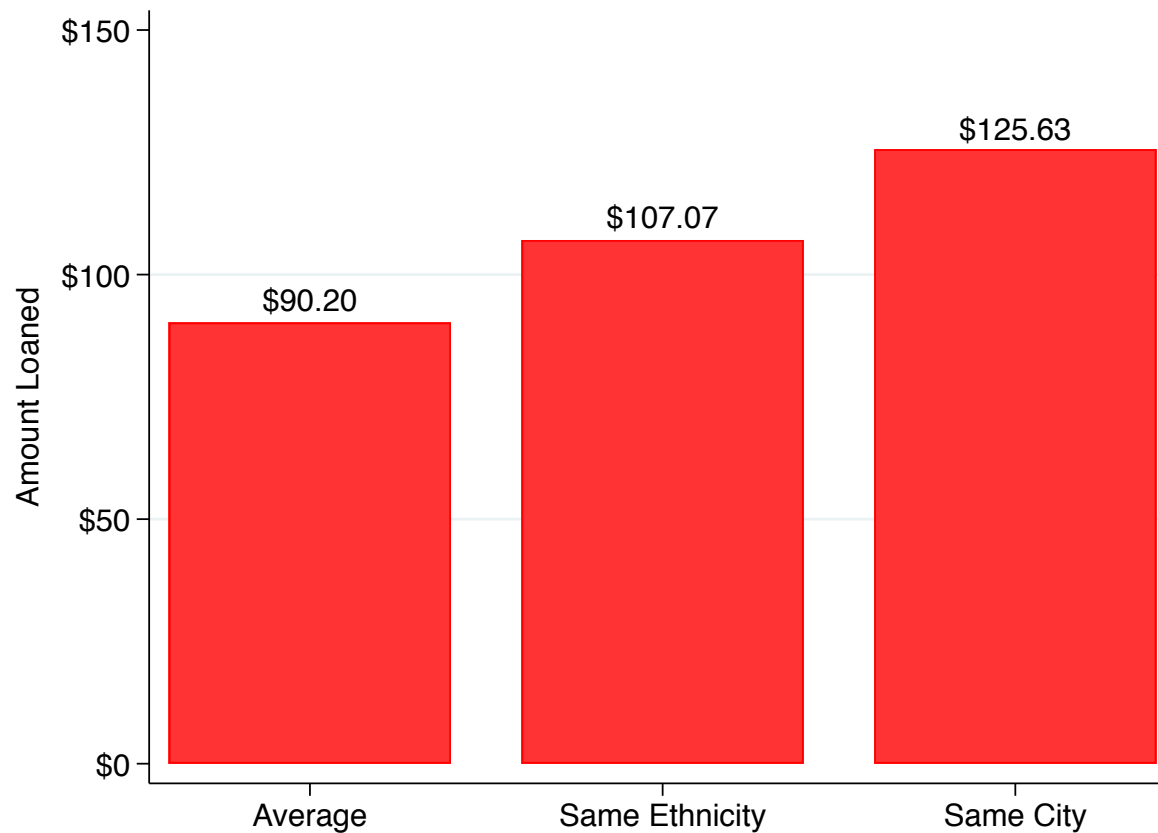
Study: University of Zurich students, donation to student funds (Frey and Meier, 2004)

- (High Giving Norm) 64% of students donate
- (Low Giving Norm) 46% of students donate



Liking

Study: Loans through peer-to-peer lending platform (e.g., Prosper.com) (Ravina, 2008)



First Day Survey

- What three things would you most like to improve about your decision making?

First Day Survey

Theme	Examples	%	
Framework		30%	
	- frameworks		10%
	- processes		16%
	- strategies		13%
Information		50%	
	- know what information is relevant		6%
	- gather information more effectively		15%
	- data-driven processes		31%
	- understanding relevance of info		18%
Organization		22%	
	- stakeholder management		12%
	- managing teams		4%
	- communicating decisions		6%
	- organization systems		6%
Timely		49%	
	- make quicker decisions		25%
	- devote appropriate amount of time		15%
	- deal with indecisions		8%
Confidence		25%	
	- trusting judgment		21%
	- overcoming fear		6%
	- communicate confidently		4%
Risk		19%	
	- take more risks		10%
	- reduce reckless decision making		9%
Learning	- learn from past decisions	15%	

Two Standard Complaints

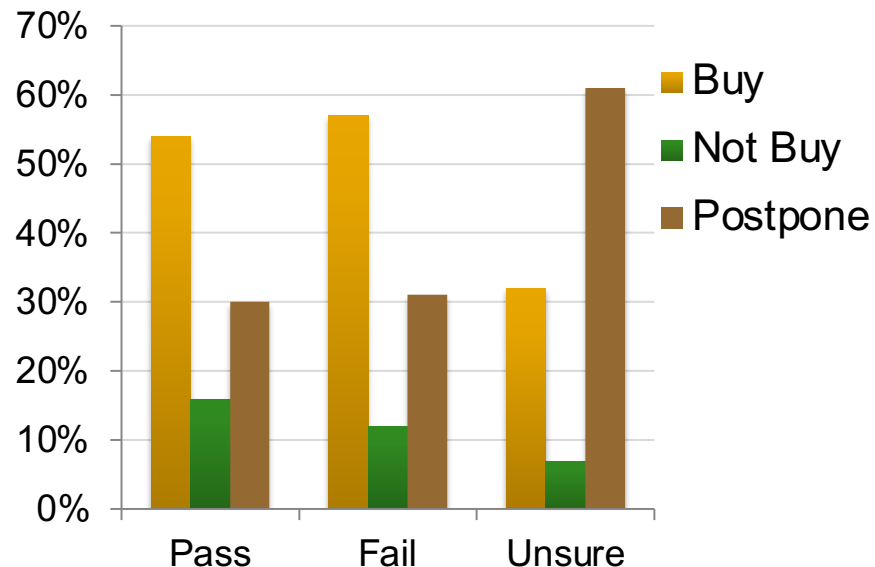
- Decisions are made too slowly
 - Conflict disables decision making
 - We acquire more information when we are conflicted, even if the information isn't necessarily going to help us resolve the conflict.
 - Prescription: Be cognizant of the costs and benefits of acquiring information (more on *Value of Information*)
 - Making Decisions $\neq$ Taking an exam
 - Information is sufficient if it enables you to make a decision
 - Put in only as much work as necessary
- Managers are too risk-averse
 - Failure >> Success (Organizational loss aversion)
 - Hindsight bias
 - Decisions are evaluated/made in isolation

Smarter decision making means better and faster

How uncertainty disables choices

- Tversky and Shafir (1992)

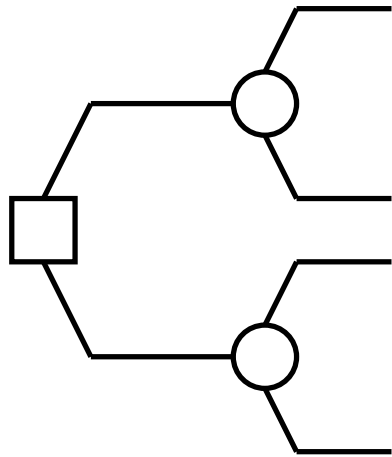
- You are a college student who has just taken a tough qualifying exam and ...
 - Are not sure if you have passed or failed
 - Passed the exam
 - Failed the exam
- You have the opportunity to buy a attractively-priced vacation, but you can postpone the choice until tomorrow for \$5.



Dealing with Analysis Paralysis

- Three prescriptions
 - Budget time for each decision
 - Be explicit about what you will learn and how you will behave differently based on what you learn
 - Identify the costs of waiting
-

Decision Trees: Garbage In, Garbage Out



■ Alternatives

- ❑ Decision tree only includes identified alternatives (Nutt [1993] study of 168 corporate decisions; 28% had one option)

■ Beliefs

- ❑ What if the probabilities are systematically biased/noisy?
- ❑ What if you fail to identify a critical uncertainty?

■ Consequences

- ❑ How do/should you measure/weigh the outcomes?
- ❑ What if you fail to consider an important objective?

A Decision Making Process I

■ Step 1: Structure Problem (ABCs)

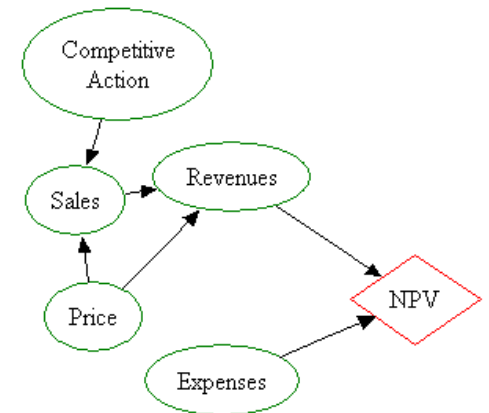
- A) Generate a set of alternatives
 - Use *strategy tables* (Howard, 1988)
- B) Identify critical uncertainties
 - Use an *influence diagram* to identify the major uncertainties (Strategic Decisions Group)
- C) Identify (and prioritize) critical objectives, including appropriate risk attitudes (as in Negotiations)

Note 1: Determine the appropriate amount of resources [time, people, etc.] to devote to this decision.

Note 2: Be clear about what is and is not included in analysis.

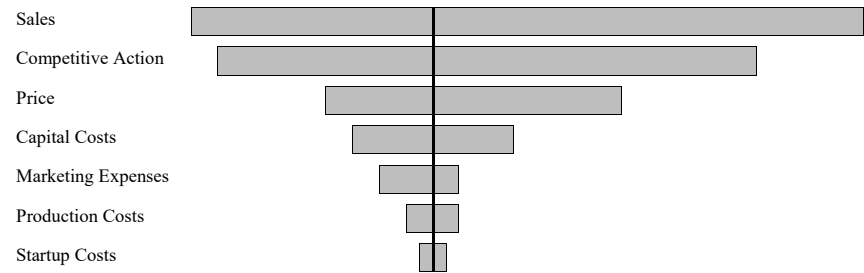
■ Step 2: Quantify uncertainties (preliminary):

- Low, Medium, and High Case [.05, .50, .95 fractiles; Class 2]; take lowest of low and highest of high



A Decision Making Process II

- Step 3: Rank uncertainties in order of importance (*Tornado Chart*)



- Step 4: Refine estimates of “critical” uncertainties
 - Identify and repair decision traps that may lead to poor estimates
 - See Cognitive Repairs
 - See prescriptions for Class 2 (esp. contrarian)
 - Recognize the tendency to collect confirmatory evidence (and interpret ambiguous evidence as consistent with existing beliefs)
 - Conduct pre-mortem

A Decision Making Process III

- Step 5: Collect information on “critical” uncertainties if value of information exceeds cost of collecting information (*Value of Information*)
- Step 6: Choose the best alternative (*Decision Analysis and Decision Trees* [Class 1 note])
 - Use sensitivity analysis if needed to increase confidence in decision
 - Make sure that you are taking the appropriate amount of risk.
- Step 7: Communicate the decision to the relevant stakeholders. Good decision makers are multi-lingual.

A Decision Making Process IV

- Step 8: Manage and monitor “critical” risks and opportunities identified by scenarios
- Step 9: Document process and learn from mistakes.

Note: A good decision process should be seen as a process for generating insight. Accordingly, steps are skipped/repeated depending on what you learn.

Influence Diagram Basics

■ Basic Purpose

- ❑ Structured approach for identifying the *uncertainties* that influence the outcome (objectives)

■ Components

- ❑ Objective 
- ❑ Decision 
- ❑ Uncertainty 
- ❑ Calculation  *determined by the predecessor nodes*

■ How to construct an influence diagram

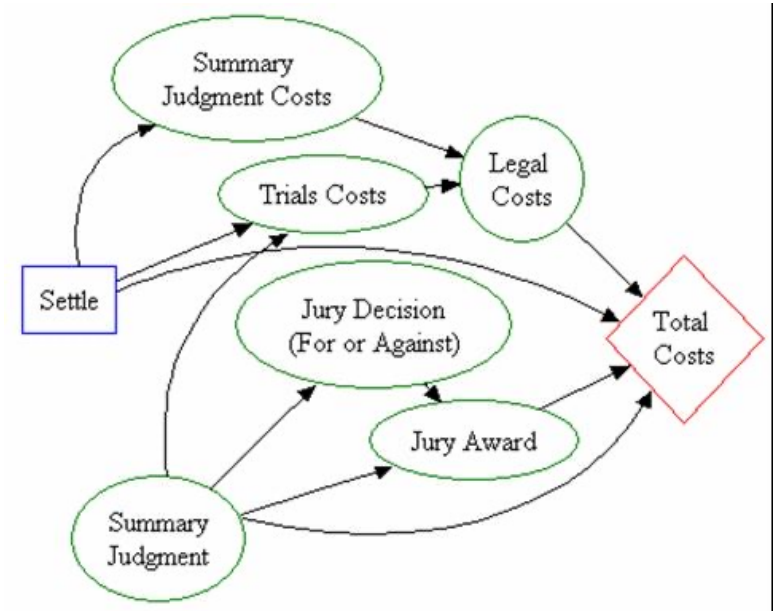
- ❑ Start with objective (*Clairvoyance Test*)
- ❑ Ask:
 - “what influences this objective?” (*Clairvoyance Test*)
 - “does anything else influence this objective?”
- ❑ Repeat with predecessors:
 - “what influences this uncertainty?” (*Clairvoyance Test*)
 - “does anything else influence this uncertainty?”

Influence Diagram Example

Sarah Gable is the President of Gable's Department Store. Last year, Bob Smith, a 35 year-old man fell on the ice in the Gable department store parking lot. Smith hurt his wrist and fractured his ankle in the accident and has brought a suit against Gable's.

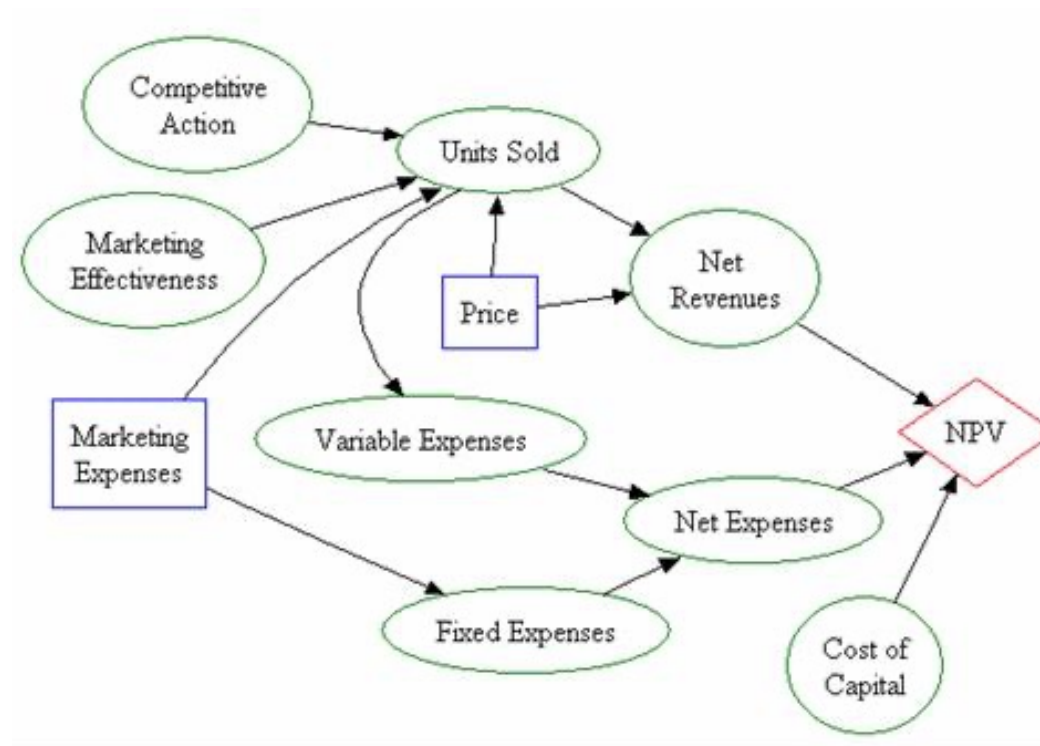
Of course, Sarah Gable would like to settle out of court and hence has initiated discussion with Smith's counsel. Although Smith's attorney is willing to settle for \$50,000, he has made it clear that if this offer is not accepted then he is willing to go to court.

The first step of the process is a summary judgment in which the judge will decide whether to hear the case. It will cost Gable \$3,000 to \$5,000 in attorney's fees to prepare for summary judgment, and she estimates that there is 90% chance that the judge will decide to hear the case and a 10% chance that the case will be dismissed. If the case is heard, Gable's best guess is that there is a 2-in-3 chance that Smith will win, with an award of anywhere from \$25,000 to \$150,000. A court trial will cost Gable another \$10,000 to \$15,000 in lawyer's fees.



Generic Financial Influence Diagram

A good part of the influence diagram in “basic” business decisions follows from the income statement.

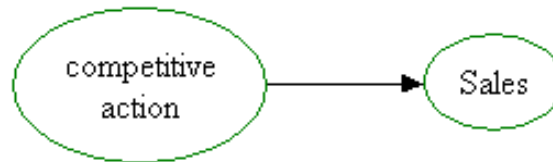


For Reference Only!

Tornado Charts

■ Step 1

- For each uncertainty (not determined by predecessors):
 - assess “low”, “medium” (“base”), “high” case [corresponding to .05, .50, and .95 fractiles]
- For uncertainties *conditionally influenced* by predecessor, assess *conditional fractiles*
 - Example:



- .05, .50, .95 fractiles for sales, conditional on “low” competitive action
- .05, .50, .95 fractiles for sales, conditional on “medium” competitive action
- .05, .50, .95 fractiles for sales, conditional on “high” competitive action

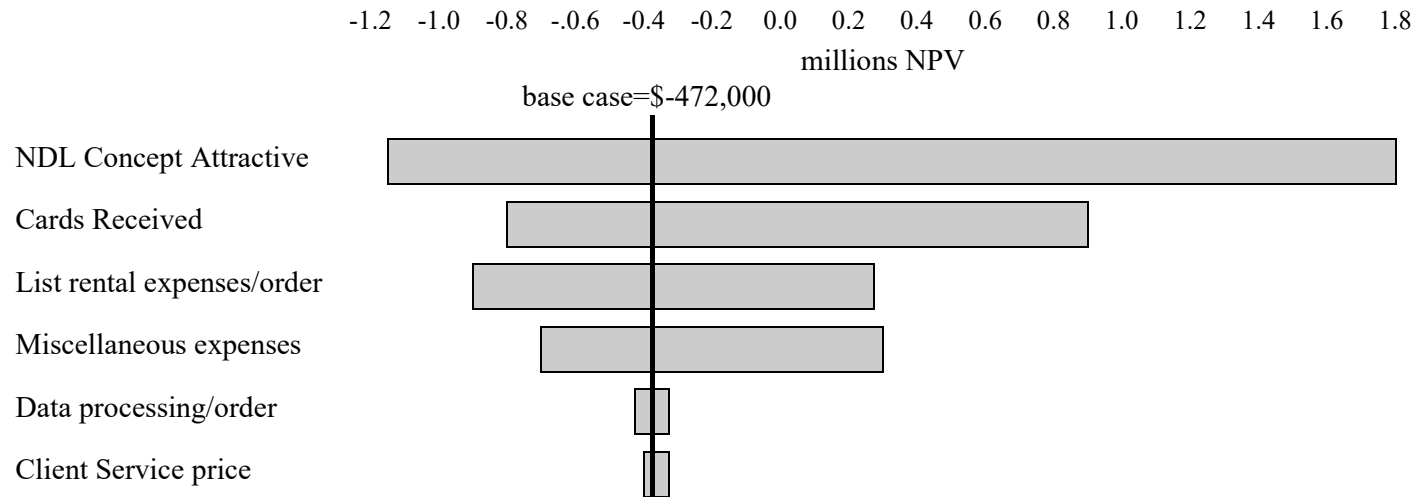
■ Step 2

- Develop a *deterministic* model
- Run all values at base case [.50 fractile] $\Rightarrow$ base case NPV

■ Step 3

- For each variable (e.g., demand):
 - change demand to low (keeping all other variables at their base case) $\Rightarrow$ low NPV
 - change demand to high (keeping all other variables at their base case) $\Rightarrow$ high NPV
- Plot all uncertainties in order of variance (high NPV minus low NPV)

Representative Tornado Chart



■ Notes

- ❑ a tornado chart is a type of *sensitivity analysis*
- ❑ “80-20 rule”: most of the variance in the objective is typically caused by 2 or 3 variables → other variables can be ignored

■ Caveats

- ❑ Positive correlation between variables (scenario analysis)
- ❑ Multiplicative effects (e.g., price and sales)

Valuing Information I

■ Basic Question

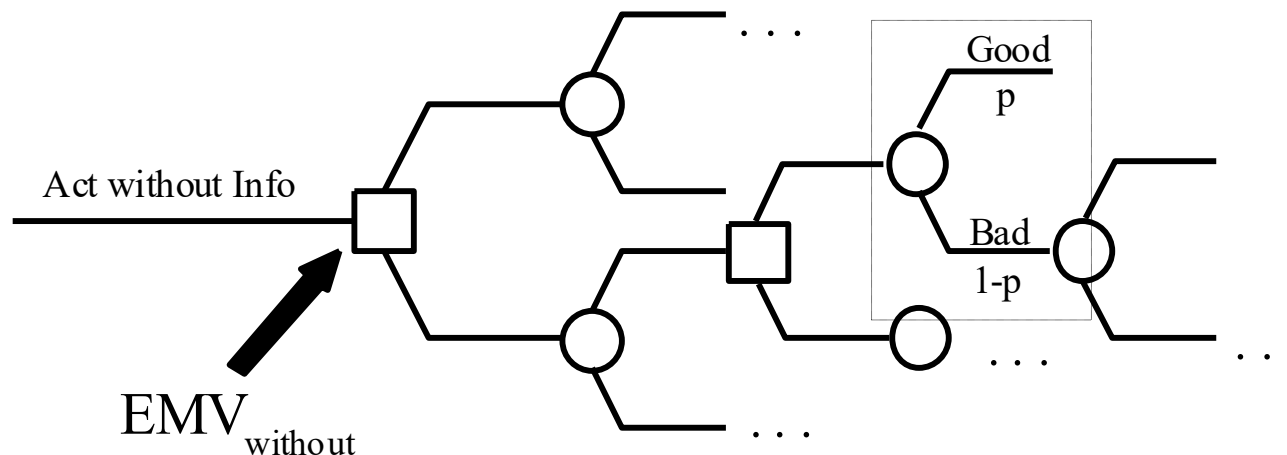
- ❑ What is the *most* I would pay to completely (incompletely) resolve my uncertainty about some uncertain quantity (demand; weather; technological success; reservation price; etc.)?
- ❑ Positive value of information results from being able to coordinate your decision with the information received and hence avoid *regret*. *Information is only worthwhile if you will change your decision based on the information.*

■ Simplifying assumptions evoked

- ❑ Information is perfect (uncertainty is completely resolved)
- ❑ Expected value maximization
 - More complicated versions are available

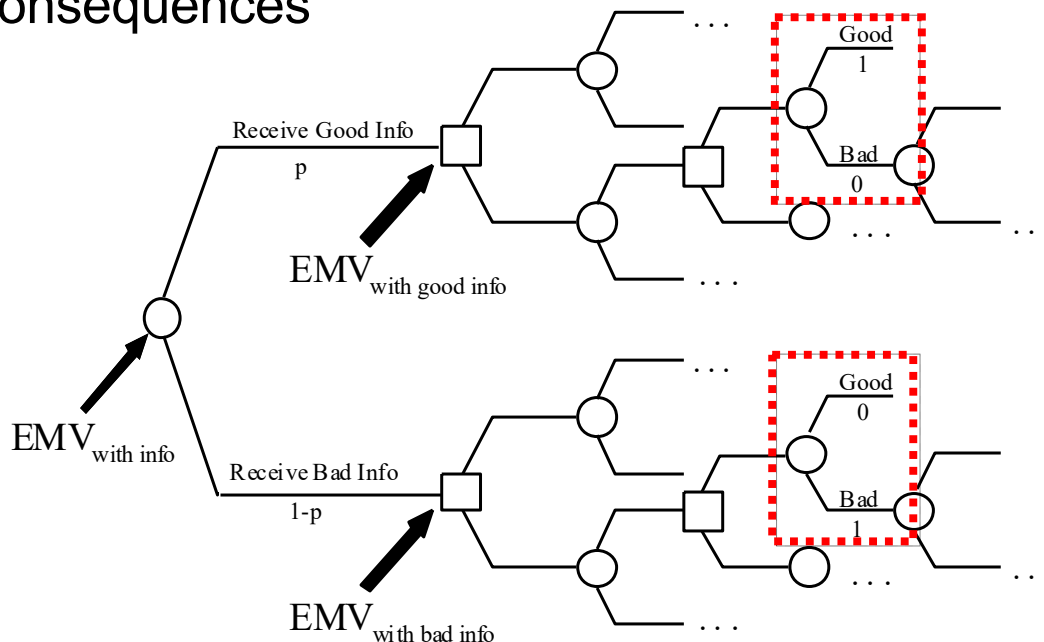
Valuing Information II

- Calculating value of (perfect) information
 - Calculate expected value assuming chronology without information (A):
 - Make decision
 - Uncertainty resolved
 - Consequences



Valuing Information III

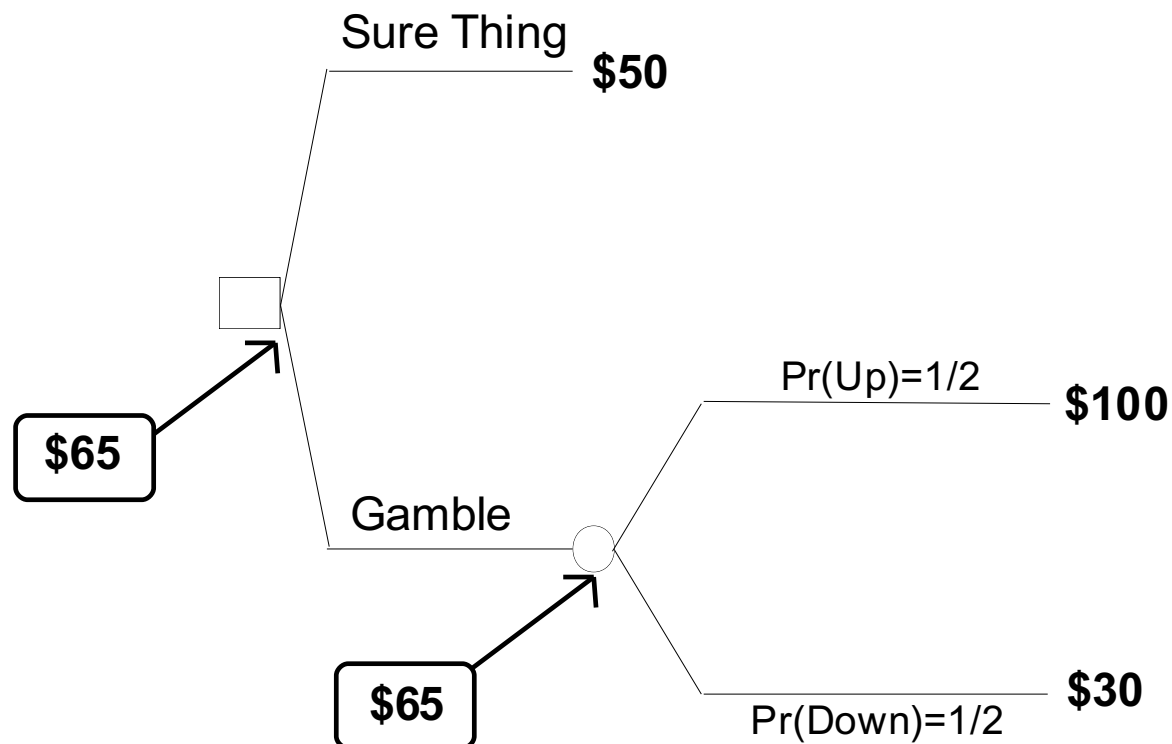
- Calculate expected value assuming chronology with information, i.e., assuming uncertainty is resolved before you need to choose (B):
 - Uncertainty resolved
 - Make decision (contingent on information)
 - Consequences



For Reference Only!

Value of Information Example I

- You can choose between \$50 for sure, or a gamble that pays \$100 if the market is up and \$30 if the market is down. (The market is equally likely to go up or down.) What would you do?

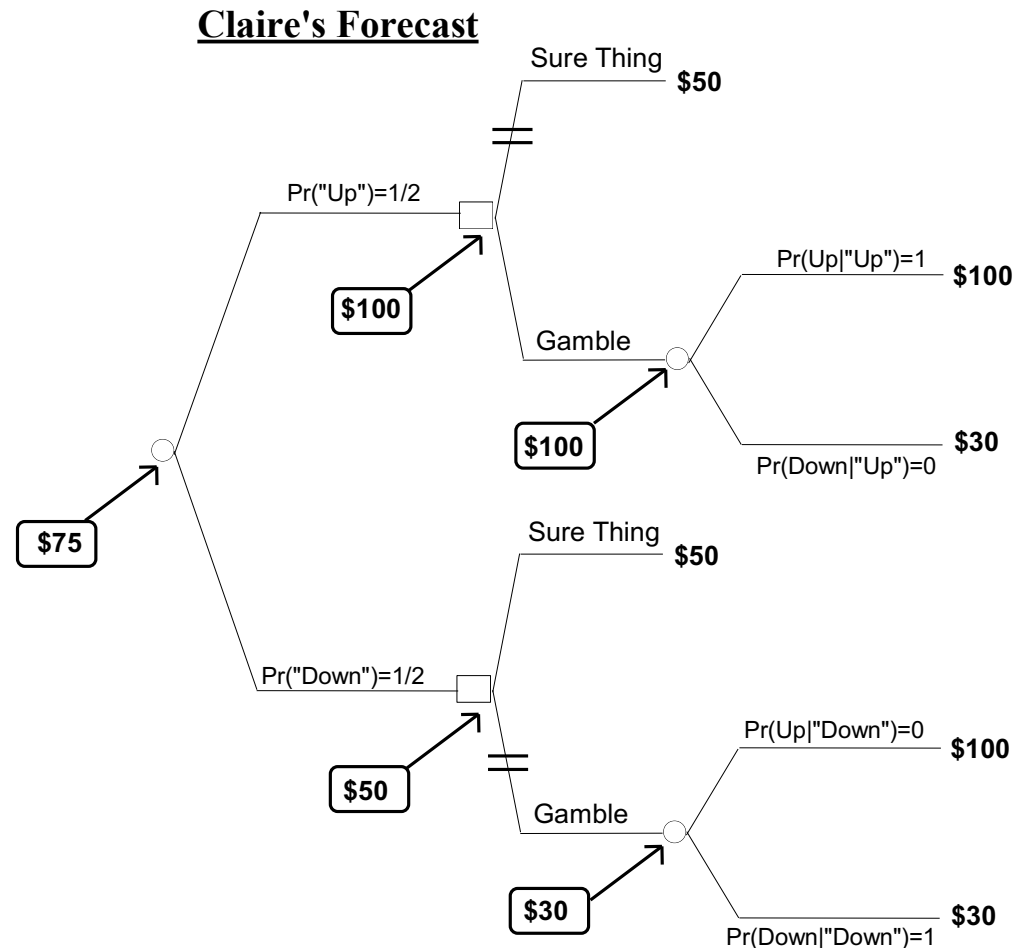


For Reference Only!

Value of Information Example II

- Suppose that a clairvoyant (Claire) can perfectly resolve the uncertainty about the market? What is the most that you would pay for this help?

$$\text{EVPI} = \$10 = \text{EV with info} - \text{EV w/o info}$$



A Harder Example

ABC company is under contract to make 10,000 blenders. It must decide whether to make or buy the blender motor. Unfortunately, ABC does not presently have motor technology. Motor development is a two-stage process. The first stage has a 80% chance of success and will cost \$30,000. The second stage has a 60% chance of success and will cost \$20,000. If development is successful, then the variable cost of producing a motor will be \$2.50/motor. If either development stage is unsuccessful, ABC must purchase motors outside at a cost of \$10/motor. Should ABC purchase the motor outside or begin development?

Answers found on Chalk under Class 5 handouts.

A Harder Example II

What is the most ABC would pay to know, before making the first decision, whether the first stage would succeed? the second stage (assume that you can resolve the uncertainty about the second stage without resolving the uncertainty about the first stage, i.e., they are separate processes)?

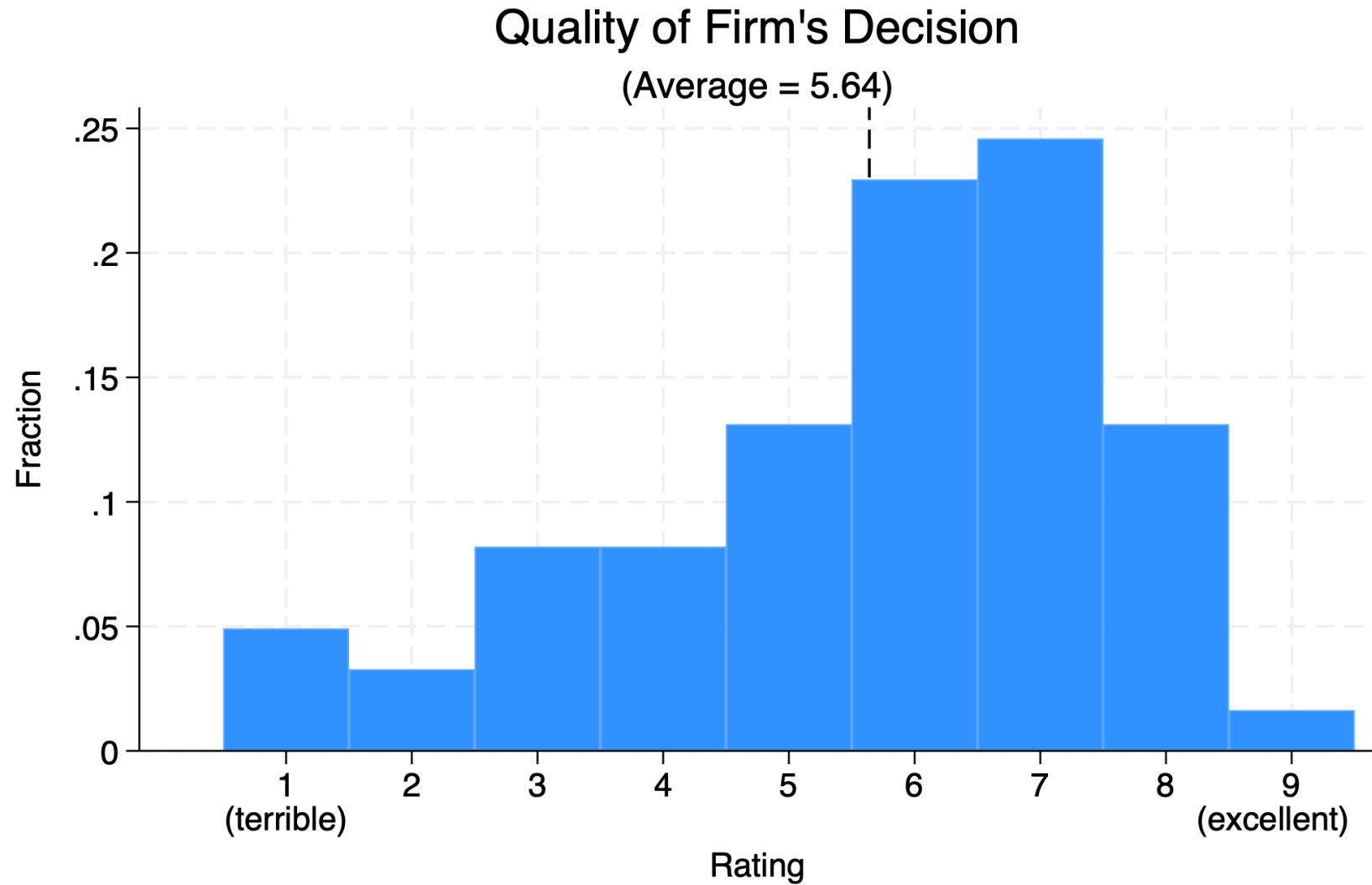
Answers found on Chalk under Class 5 handouts.

For Reference Only!

Decision Analysis “Hints”

- Consider only outcomes that differ across actions
 - Example: For a cost minimization decision, no need to consider revenue
- Model decision tree according to order of actions and resolution of uncertainties
 - Make sure that critical potential actions are not omitted (opportunities to collect information, stage investments, cancel or increase commitments, etc.)
- Build minimally complicated decision tree
 - Model most important elements initially. Is insight sufficient to make a decision? If not, consider secondary elements. Iterate as needed.
 - Suppose insights are sufficient to make a decision. Can you do even better by managing risk, or eliminating shortcomings of action?

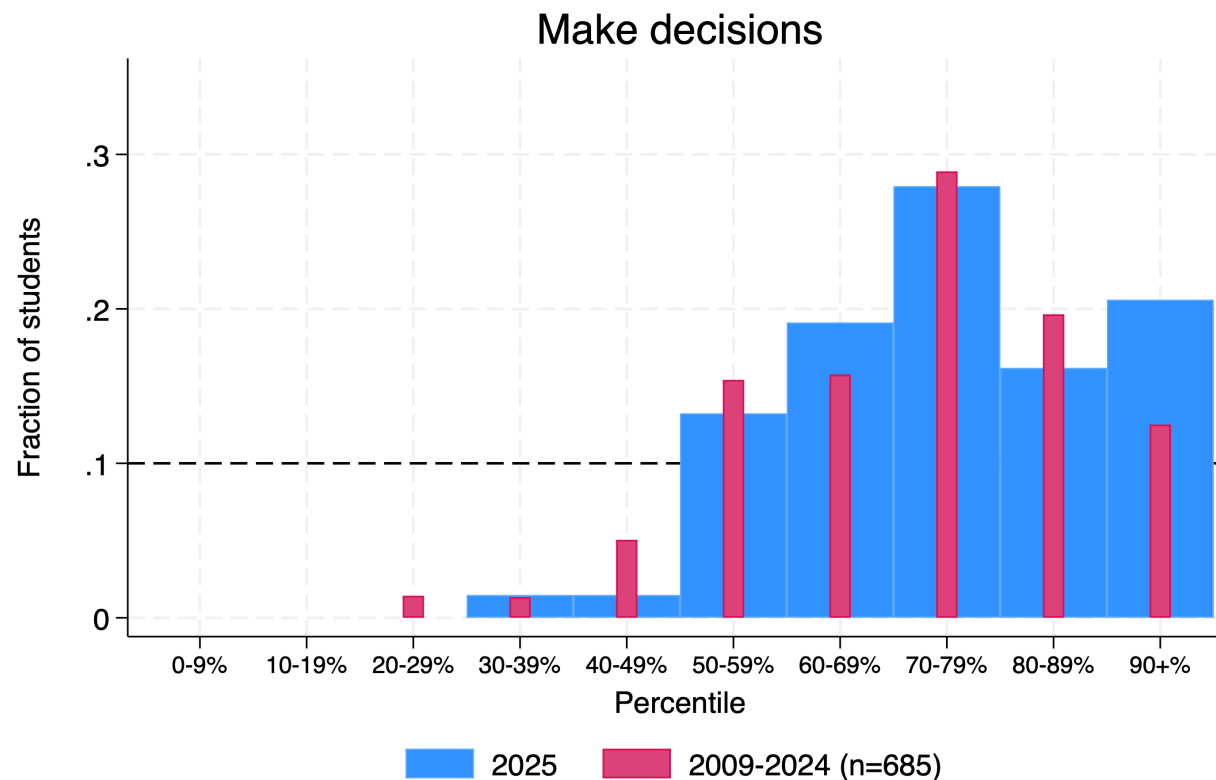
Decision Quality Assessment



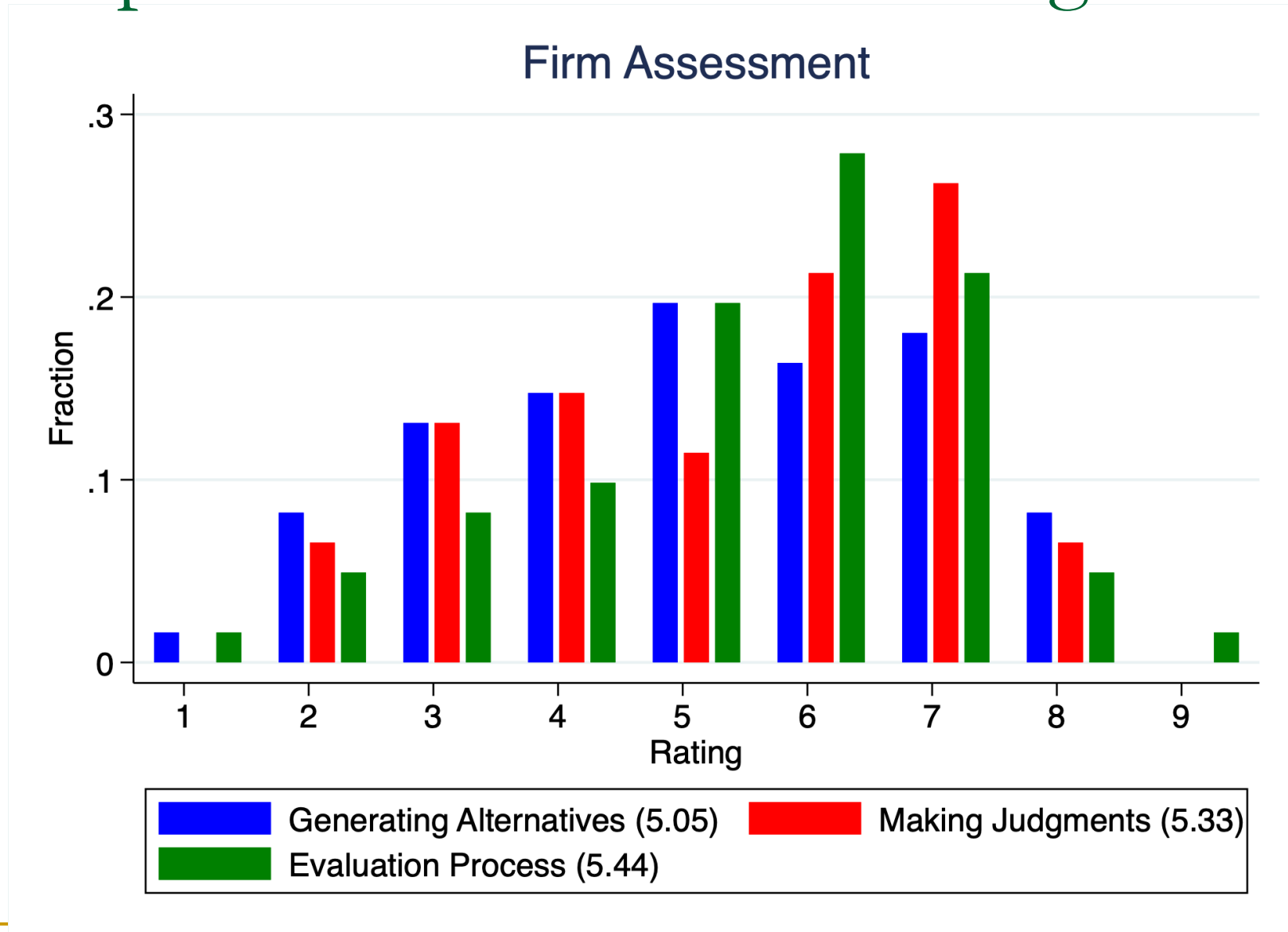
Bonus Example II

Question:

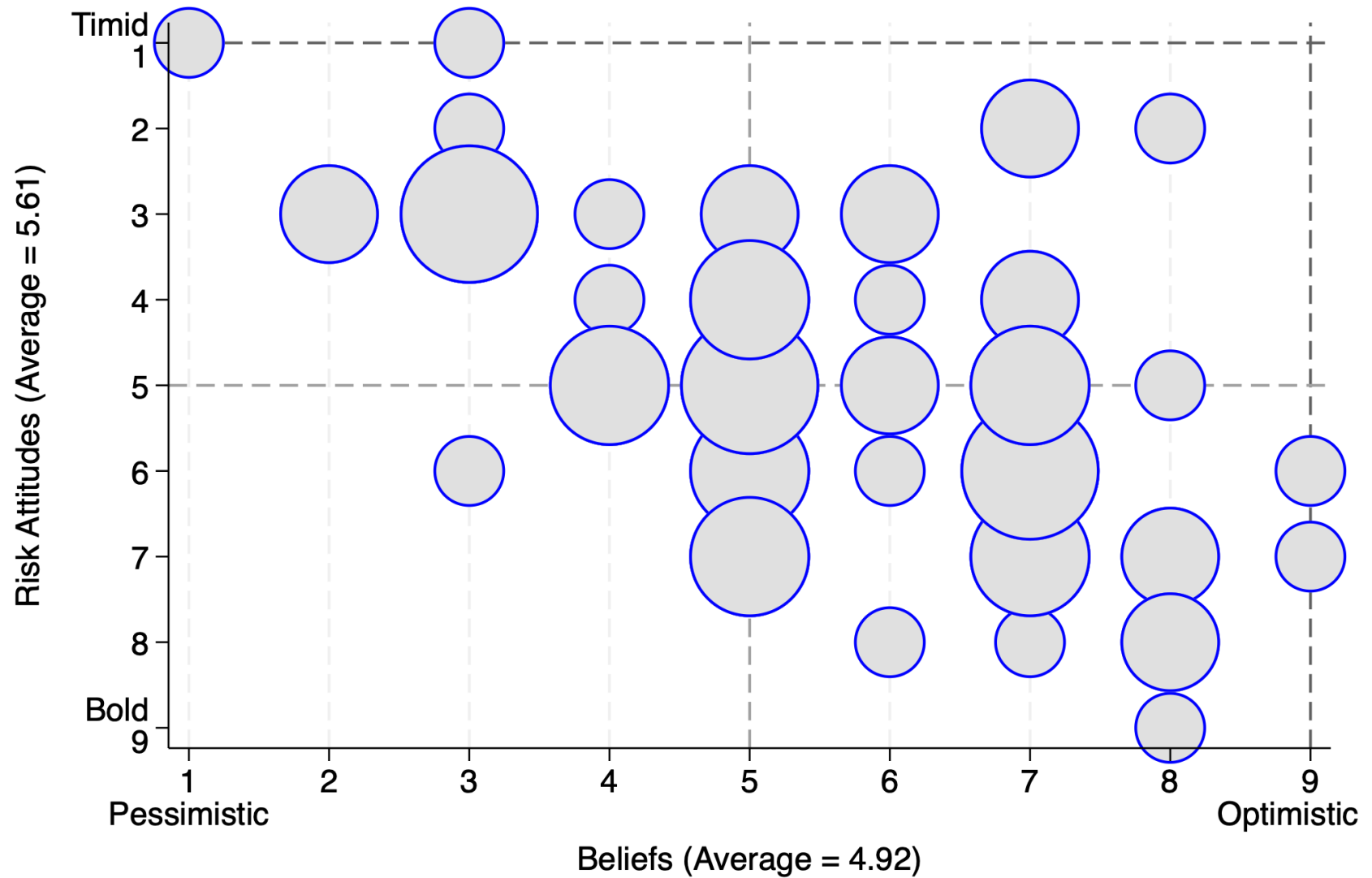
- Compared to the other students in this class, how do you think you rate in ability to make decisions?



Components of Decision Making



Beliefs versus Risk Attitudes



Your Firms: “Best Practices” I

- Formal Processes

- Decision Frameworks

- Collecting, processing, and evaluating information
 - Knowledge management
 - Probability assessment (data driven)
 - Long-term forecasts and planning

- Identifying alternatives

- Evaluation (clear objectives, sometimes about risk)

- Fast!

- Scenario Analysis

- “Consensus with qualification”

- Learning

- Record Keeping

- Post-mortem/Audit processes

- Evaluation processes

- Little penalty for failure

Your Firms: “Best Practices” II

- Implementation
 - ❑ Clear communications of decision
- “Cognitive Repairs”
 - ❑ Benchmarking
 - ❑ Multiple opinions
 - ❑ External opinions
 - ❑ Independent analysis
 - ❑ Evolution of fast processes
 - ❑ Built-in Contrarian
 - ❑ Cross-functional teams
 - ❑ Formal rotations

Your Firms: “Worst Practices” I

■ Decision Processes

- ❑ None
- ❑ Risk-averse
- ❑ Slow
- ❑ Reactive not proactive
- ❑ Analysis paralysis
- ❑ Inconsistent
- ❑ One-size fits all
- ❑ Uncertainty ignored
- ❑ Unclear objectives
- ❑ Always do what was done in the past

■ Implementation

- ❑ Decisions poorly communicated

Your Firms: “Worst Practices” II

■ Organizational

- ❑ Incentives
- ❑ Evaluation
- ❑ No one will challenge the boss
- ❑ Politics
- ❑ Too many layers of bureaucracy
- ❑ Overly parochial
- ❑ Consensus Processes (Autocratic, Need for Unanimity, etc.)

■ Psychological

- ❑ Confirmatory reasoning (no questioning of assumptions)
- ❑ Emotion over analysis
- ❑ Anecdotes over data
- ❑ Hindsight bias
- ❑ Anchor-based forecasts
- ❑ Inside view (Optimistic)
- ❑ No learning from mistakes

Three Approaches to Organizations

- Fast Decision Making

- Eisenhardt (1990)

- Cognitive Repairs

- Heath, Larrick, and Klayman (1998)

- Re-engineering Decision Processes

- Smithkline Beecham, *Harvard Business Review* (1998)

Accelerating Decision Making

■ Study

- ❑ 12 microcomputer firms
- ❑ Decisions: products, alliances, strategy
- ❑ “Fast decision makers”: 2 to 4 months
- ❑ “Slow decision makers”: 12 to 18 months

■ Findings: What do “fast” firms do?

- ❑ Track real time information (slow firms focus on planning or “futuristic” information)
- ❑ Consider multiple alternatives
- ❑ Use experienced counselors (often internal)
- ❑ “Consensus with qualification”
- ❑ Integration of decisions

■ Do these make sense?

Cognitive Repairs

- Source

- ❑ Heath, Larrick, Klayman (1998)

- What is a cognitive repair?

- ❑ Decision biases often result from shortcoming in human judgment (cognition)
 - ❑ Organizations evolve practices of repairing these shortcomings
 - ❑ Information collection/gathering vs. Information weighting

- Tradeoffs

- ❑ simple versus complex (ease of use vs. accuracy)
 - ❑ domain-specific versus domain-general
 - ❑ corrective versus preventative (corrective can be hard because it is difficult to “unlearn”)
 - ❑ familiar versus novel
 - ❑ social versus individual

Re-engineering Decision Processes

■ Examples

- ❑ Smithkline Beecham
- ❑ General Motors (Decision Dialogue Process: dialogue between decision making team and cross-functional decision team)

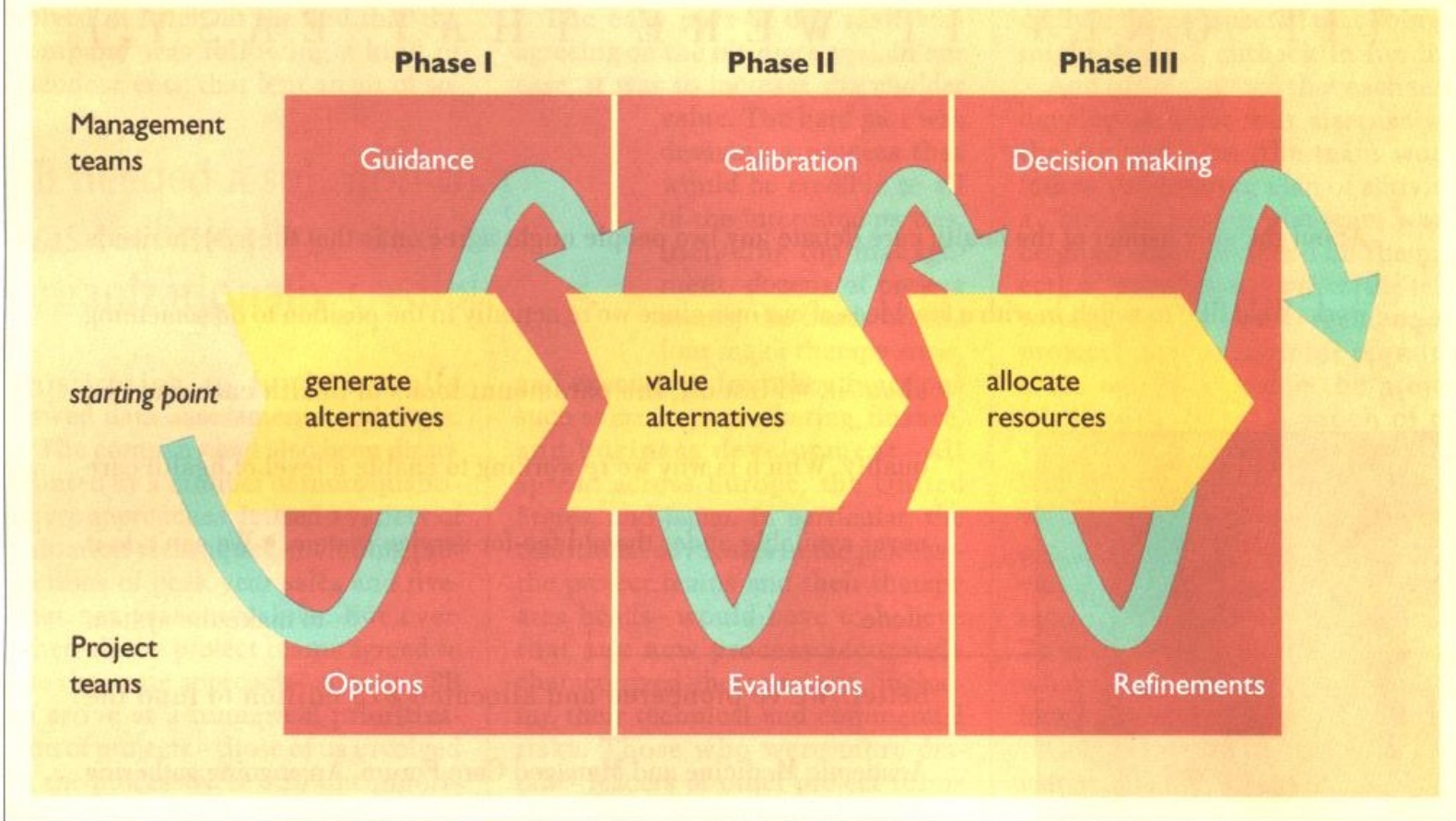
■ What do these processes do?

- ❑ Establishes common language
- ❑ Establishes constant dialogue between decision makers and decision implementers
- ❑ Uses structured process
- ❑ Deals with organizational barriers

The SB Process

A FRAMEWORK FOR EFFECTIVE COMMUNICATION

This three-phase process allows SB to find more value in its portfolio of development projects.



Good Organizational Decision Process

- Logical
- Structured
- Transparent
- Consistent, but scaleable
- Participatory, but not consensus bound
- Minimizes psychological biases
- Sensitive to organizational and cultural issues

What can you do?

- Three Different Approaches

- ❑ Rebuilding/reforming decision processes
- ❑ Organizational Repairs
- ❑ Managing others
 - Your boss
 - Your peers/work team
 - Your subordinates



For each of the “others”, identify a way in which you may “manage” that person to improve your organization’s decision making

Wu's Clues: Part 5



Clue 18: Choosing is hard. Help people out by norming, reminding, simplifying, defaulting, and helping them pre-commit.

Clue 19: You have to think *smarter* to think less.

Clue 20: Think of a manager as a decision facilitator instead of a decision maker.

Clue 21: When confused, think back to your ABCs. (Alternatives, Beliefs, and Consequences).

	Decision Traps	Prescriptions
Alternatives	• Addressing the incorrect problem • Insufficient generation of alternatives • Ignoring contingencies/Future strategies	• Understanding your objectives (Negotiations) • Expanding w/ creativity (you) • Limiting (others) • Choice architecture • Scenario Analysis/Risk Management (*)
Beliefs	• Focusing on all uncertainties equally • Overconfidence (narrow confidence intervals) • Excessive optimism • Anchoring and availability • Hindsight bias • Confirmation bias • Failure to learn over time	• Knowing sources → Cognitive Repairs • Feedback • Disconfirming questions/contrarian • Culture of learning • Scenario Analysis (*) • Value of Information (*)
Consequences	• Susceptibility to framing • Insufficient risk taking (loss aversion/regret/hindsight bias) • Overweighting hard or soft • Relying only on intuition	• Evaluating based on process, not outcome • Portfolio approach • Being rigorous about soft issues, as well as hard issues